

REACT: SYNERGIZING REASONING AND ACTING IN LANGUAGE MODELS

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ABSTRACT

While large language models (LLMs) have demonstrated impressive performance across tasks in language understanding and interactive decision making, their abilities for reasoning (e.g. chain-of-thought prompting) and acting (e.g. action plan generation) have primarily been studied as separate topics. In this paper, we explore the use of LLMs to generate both reasoning traces and task-specific actions in an interleaved manner, allowing for greater synergy between the two: reasoning traces help the model induce, track, and update action plans as well as handle exceptions, while actions allow it to interface with and gather additional information from external sources such as knowledge bases or environments. We apply our approach, named ReAct, to a diverse set of language and decision making tasks and demonstrate its effectiveness over state-of-the-art baselines in addition to improved human interpretability and trustworthiness. Concretely, on question answering (HotpotQA) and fact verification (Fever), ReAct overcomes prevalent issues of hallucination and error propagation in chain-of-thought reasoning by interacting with a simple Wikipedia API, and generating human-like task-solving trajectories that are more interpretable than baselines without reasoning traces. Furthermore, on two interactive decision making benchmarks (ALFWorld and WebShop), ReAct outperforms imitation and reinforcement learning methods by an absolute success rate of 34% and 10% respectively, while being prompted with only one or two in-context examples.

1 INTRODUCTION

A unique feature of human intelligence is the ability to seamlessly combine task-oriented actions with verbal reasoning (or inner speech, Alderson-Day & Fernyhough, 2015), which has been theorized to play an important role in human cognition for enabling self-regulation or strategization (Vygotsky, 1987; Luria, 1965; Fernyhough, 2010) and maintaining a working memory (Baddeley, 1992). Consider the example of cooking up a dish in the kitchen. Between any two specific actions, we may reason in language in order to track progress (“now that everything is cut, I should heat up the pot of water”), to handle exceptions or adjust the plan according to the situation (“I don’t have salt, so let me use soy sauce and pepper instead”), and to realize when external information is needed (“how do I prepare dough? Let me search on the Internet”). We may also act (open a cookbook to read the recipe, open the fridge, check ingredients) to support the reasoning and to answer questions (“What dish can I make right now?”). This tight synergy between “acting” and “reasoning” allows humans to learn new tasks quickly and perform robust decision making or reasoning, even under previously unseen circumstances or facing information uncertainties.

Recent results have hinted at the possibility of combining verbal reasoning with interactive decision making in autonomous systems. On one hand, properly prompted large language models (LLMs) have demonstrated emergent capabilities to carry out several steps of reasoning traces to derive

^{*}Work during Google internship. Projct page with code: <https://react-lm.github.io/>.

R E A C T : 在语言模型中协同推理与行动

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摘要

虽然大型语言模型（LLMs）在语言理解和交互式决策等任务上表现出色，但它们在推理（例如思维链提示）和行动（例如行动计划生成）方面的能力主要作为独立主题进行过研究。在本文中，我们探索了使用LLMs以交错方式生成推理轨迹和特定任务行动的方法，从而实现两者之间的更大协同：推理轨迹帮助模型诱导、跟踪和更新行动计划以及处理异常，而行动则允许模型与外部来源（如知识库或环境）交互并获取更多信息。我们将名为ReAct的方法应用于一系列语言和决策任务，并证明其在最先进基线之外的有效性，同时还提高了人类可解释性和可信度。具体而言，在问答（HotpotQA）和事实验证（Fever）上，ReAct通过与简单的维基百科API交互，并生成比没有推理轨迹的基线更具人类样式的任务解决轨迹，从而克服了思维链推理中常见的幻觉和错误传播问题。此外，在两个交互式决策基准（ALFWorld和WebShop）上，ReAct分别比模仿学习和强化学习方法绝对成功率高出34%和10%，而只需一个或两个上下文示例提示。

1 简介

人类智能的独特之处在于能够无缝地将面向任务的动作与语言推理（或内部言语, Alderson-Day & Fernyhough, 2015）相结合，这种能力被认为在人类认知中起着重要作用，能够实现自我调节或策略制定 (Vygotsky, 1987; Luria, 1965; Fernyhough, 2010)，并维持工作记忆 (Baddeley, 1992)。以在厨房里做一道菜为例。在任何两个特定动作之间，我们可能会用语言进行推理，以跟踪进度（“现在一切都切好了，我应该加热锅里的水”），处理异常或根据情况调整计划（“我没有盐，所以让我用酱油和胡椒代替”），以及意识到何时需要外部信息（“我该怎么准备面团？让我上网搜索一下”）。我们还可以采取行动（打开食谱书阅读菜谱、打开冰箱、检查食材）来支持推理并回答问题（“我现在能做什么菜？”）。这种“行动”与“推理”之间的紧密协同作用，使人类能够快速学习新任务，并在之前未见过的环境或面对信息不确定的情况下进行稳健的决策或推理。

近期的研究表明，在自主系统中结合语言推理与交互式决策的可能性。一方面，经过适当提示的大型语言模型（LLMs）已经展现出执行推理轨迹多步骤以推导出

^{*}在谷歌实习期间工作。项目页面带代码：<https://react-lm.github.io/>。

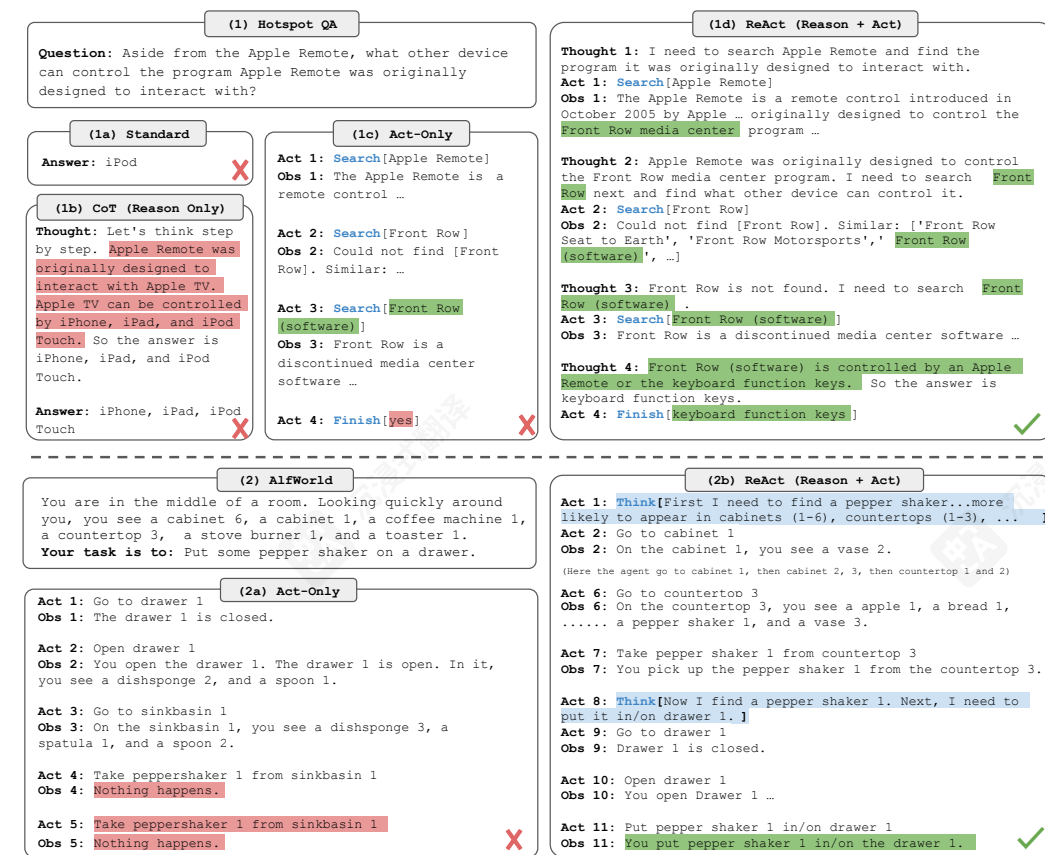


Figure 1: (1) Comparison of 4 prompting methods, (a) Standard, (b) Chain-of-thought (CoT, Reason Only), (c) Act-only, and (d) ReAct (Reason+Act), solving a HotspotQA (Yang et al., 2018) question. (2) Comparison of (a) Act-only and (b) ReAct prompting to solve an AlfWorld (Shridhar et al., 2020b) game. In both domains, we omit in-context examples in the prompt, and only show task solving trajectories generated by the model (Act, Thought) and the environment (Obs).

answers from questions in arithmetic, commonsense, and symbolic reasoning tasks (Wei et al., 2022). However, this “chain-of-thought” reasoning is a static black box, in that the model uses its own internal representations to generate thoughts and is not grounded in the external world, which limits its ability to reason reactively or update its knowledge. This can lead to issues like fact hallucination and error propagation over the reasoning process (Figure 1 (1b)). On the other hand, recent work has explored the use of pre-trained language models for planning and acting in interactive environments (Ahn et al., 2022; Nakano et al., 2021; Yao et al., 2020; Huang et al., 2022a), with a focus on predicting actions via language priors. These approaches usually convert multi-modal observations into text, use a language model to generate domain-specific actions or plans, and then use a controller to choose or execute them. However, they do not employ language models to reason abstractly about high-level goals or maintain a working memory to support acting, barring Huang et al. (2022b) who perform a limited form of verbal reasoning to reiterate spatial facts about the current state. Beyond such simple embodied tasks to interact with a few blocks, there have not been studies on how reasoning and acting can be combined in a synergistic manner for general task solving, and if such a combination can bring systematic benefits compared to reasoning or acting alone.

In this work, we present ReAct, a general paradigm to combine reasoning and acting with language models for solving diverse language reasoning and decision making tasks (Figure 1). ReAct prompts LLMs to generate both verbal reasoning traces and actions pertaining to a task in an interleaved manner, which allows the model to perform dynamic reasoning to create, maintain, and adjust high-level plans for acting (reason to act), while also interact with the external environments (e.g. Wikipedia) to incorporate additional information into reasoning (act to reason).

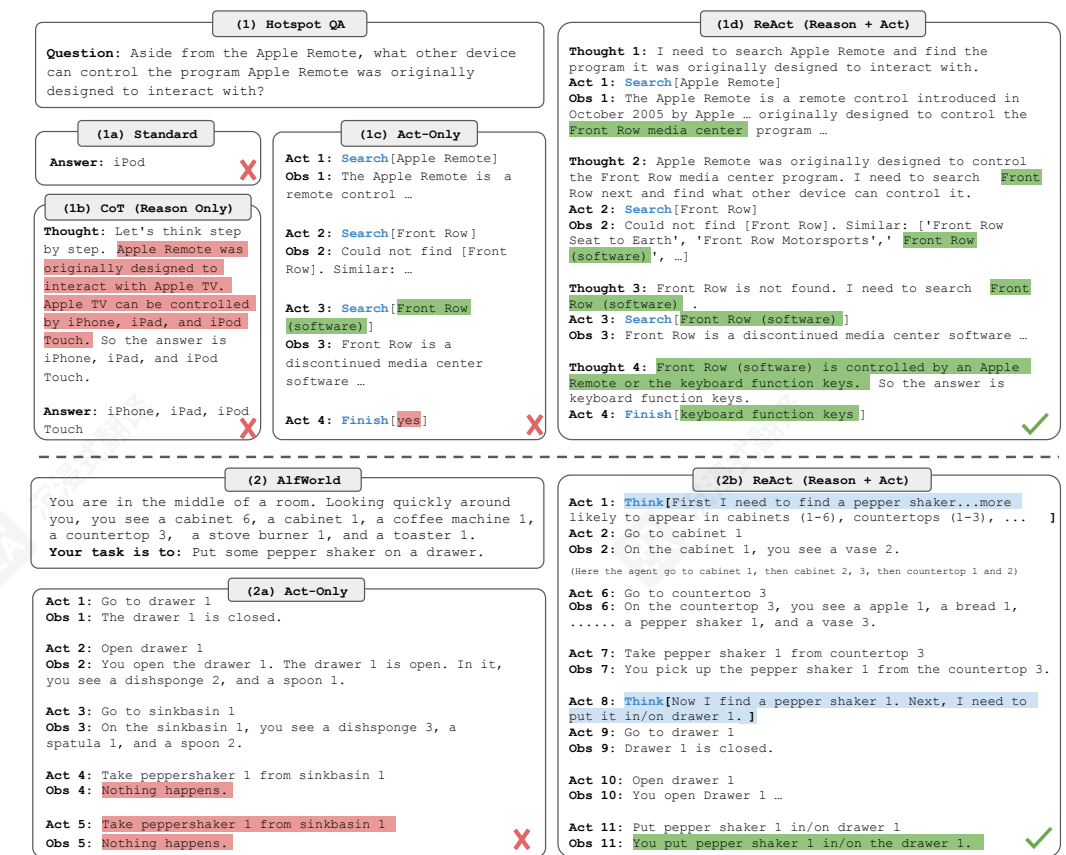


图1: (1) 4种提示方法的对比, (a) 标准, (b) 思维链 (CoT, 仅推理), (c) 仅行动, 以及 (d) ReAct (推理+行动), 解决一个HotspotQA (Yang等人, 2018) 问题。 (2) (a) 仅行动与 (b) ReAct提示方法解决一个AlfWorld (Shridhar等人, 2020b) 游戏的对比。在两个领域, 我们在提示中省略上下文示例, 仅展示模型 (行动, 推理) 和环境 (观察) 生成的任务解决轨迹。

近期研究结果已暗示了在自主系统中结合语言推理与交互式决策的可能性。一方面, 经过适当提示的大型语言模型 (LLMs) 已展现出在算术、常识和符号推理任务中 (Wei 等人, 2022) 推导出问题答案的推理能力, 能完成多个推理步骤。然而, 这种 “思维链” 推理是一种静态黑盒, 模型依赖其内部表征生成推理过程, 并未与外部世界建立联系, 这限制了其反应式推理或知识更新的能力。这可能导致事实幻觉和推理过程中的错误传播等问题 (图1 (1b))。另一方面, 近期研究探索了预训练语言模型在交互环境中的规划与行动应用 (Ahn 等人, 2022; Nakano 等人, 2021; Yao 等人, 2020; Huang 等人, 2022a), 重点通过语言先验预测行动。这些方法通常将多模态观测转换为文本, 使用语言模型生成特定领域的行动或计划, 再通过控制器选择或执行。但它们未使用语言模型进行抽象的高层目标推理或维持工作记忆以支持行动, 除Huang 等人 (2022b) 外, 后者仅执行有限的语言推理以重述当前状态的空间事实。超越与少量积木交互的简单具身任务, 目前尚无研究探讨推理与行动如何协同结合以解决一般任务, 以及这种结合相比单独推理或行动能否带来系统性优势。

在这项工作中, 我们提出了ReAct, 一种结合推理和行动的通用范式, 用于解决各种语言推理和决策任务 (图1)。ReAct以交错的方式提示大型语言模型生成与任务相关的语言推理轨迹和行动, 这允许模型进行动态推理, 以创建、维护和调整行动的高级计划 (为行动而推理), 同时还可以与外部环境 (例如维基百科) 交互, 将额外信息纳入推理 (为推理而行动)。

We conduct empirical evaluations of ReAct and state-of-the-art baselines on four diverse benchmarks: question answering (HotPotQA, Yang et al., 2018), fact verification (Fever, Thorne et al., 2018), text-based game (ALFWorld, Shridhar et al., 2020b), and webpage navigation (WebShop, Yao et al., 2022). For HotPotQA and Fever, with access to a Wikipedia API that the model can interact with, ReAct outperforms vanilla action generation models while being competitive with chain-of-thought reasoning (CoT) (Wei et al., 2022). The best approach overall is a combination of ReAct and CoT that allows for the use of both internal knowledge and externally obtained information during reasoning. On ALFWorld and WebShop, two or even one-shot ReAct prompting is able to outperform imitation or reinforcement learning methods trained with $10^3 \sim 10^5$ task instances, with an absolute improvement of 34% and 10% in success rates respectively. We also demonstrate the importance of sparse, versatile reasoning in decision making by showing consistent advantages over controlled baselines with actions only. Besides general applicability and performance boost, the combination of reasoning and acting also contributes to model interpretability, trustworthiness, and diagnosability across all domains, as humans can readily distinguish information from model’s internal knowledge versus external environments, as well as inspect reasoning traces to understand the decision basis of model actions.

To summarize, our key contributions are the following: (1) we introduce ReAct, a novel prompt-based paradigm to synergize reasoning and acting in language models for general task solving; (2) we perform extensive experiments across diverse benchmarks to showcase the advantage of ReAct in a few-shot learning setup over prior approaches that perform either reasoning or action generation in isolation; (3) we present systematic ablations and analysis to understand the importance of acting in reasoning tasks, and reasoning in interactive tasks; (4) we analyze the limitations of ReAct under the prompting setup (i.e. limited support of reasoning and acting behaviors), and perform initial finetuning experiments showing the potential of ReAct to improve with additional training data. Scaling up ReAct to train and operate on more tasks and combining it with complementary paradigms like reinforcement learning could further unlock the potential of large language models.

2 REACT: SYNERGIZING REASONING + ACTING

Consider a general setup of an agent interacting with an environment for task solving. At time step t , an agent receives an observation $o_t \in \mathcal{O}$ from the environment and takes an action $a_t \in \mathcal{A}$ following some policy $\pi(a_t|c_t)$, where $c_t = (o_1, a_1, \dots, o_{t-1}, a_{t-1}, o_t)$ is the *context* to the agent. Learning a policy is challenging when the mapping $c_t \mapsto a_t$ is highly implicit and requires extensive computation. For example, the agent shown in Figure 1(1c) is unable to generate the correct final action (Act 4) to finish the QA task as it requires complex reasoning over the trajectory context (Question, Act 1-3, Obs 1-3). Similarly, the agent shown in Figure 1(2a) fails to comprehend from the context that sinkbasin 1 does not contain peppershaker 1, thus keep producing hallucinating actions.

The idea of ReAct is simple: we augment the agent’s action space to $\hat{\mathcal{A}} = \mathcal{A} \cup \mathcal{L}$, where $\mathcal{L}$ is the space of language. An action $\hat{a}_t \in \mathcal{L}$ in the language space, which we will refer to as a *thought* or a *reasoning trace*, does not affect the external environment, thus leading to no observation feedback. Instead, a thought $\hat{a}_t$ aims to compose useful information by reasoning over the current context c_t , and update the context $c_{t+1} = (c_t, \hat{a}_t)$ to support future reasoning or acting. As shown in Figure 1, there could be various types of useful thoughts, e.g. decomposing task goals and create action plans (2b, Act 1; 1d, Thought 1), injecting commonsense knowledge relevant to task solving (2b, Act 1), extracting important parts from observations (1d, Thought2, 4), track progress and transit action plans (2b, Act 8), handle exceptions and adjust action plans (1d, Thought 3), and so on.

However, as the language space $\mathcal{L}$ is unlimited, learning in this augmented action space is difficult and requires strong language priors. In this paper, we mainly focus on the setup where a frozen large language model, PaLM-540B (Chowdhery et al., 2022)¹, is prompted with few-shot in-context examples to generate both domain-specific actions and free-form language thoughts for task solving (Figure 1 (1d), (2b)). Each in-context example is a human trajectory of actions, thoughts, and environment observations to solve a task instance (see Appendix C). For the tasks where reasoning is of primary importance (Figure 1(1)), we alternate the generation of thoughts and actions so that the task-solving trajectory consists of multiple thought-action-observation steps. In contrast, for decision making tasks that potentially involve a large number of actions (Figure 1(2)), thoughts only need to

¹We show some GPT-3 (Brown et al., 2020) results in Appendix A.1, which outperforms PaLM-540B.

我们对ReAct和最先进的基线在四个不同的基准上进行了实证评估：问答（HotPotQA, Yang等人, 2018年）、事实验证（Fever, Thorne等人, 2018年）、基于文本的游戏（ALFWorld, Shridhar等人, 2020b）和网页导航（WebShop, Yao等人, 2022年）。对于HotPotQA和Fever, 由于模型可以与之交互的维基百科API, ReAct优于纯行动生成模型, 同时与思维链推理（CoT）（Wei等人, 2022年）具有竞争力。总体最佳方法是ReAct和CoT的组合, 它允许在推理过程中使用内部知识和外部获取的信息。在ALFWorld和WebShop上, 两轮甚至单轮ReAct提示能够优于使用 $10^3 \sim 10^5$ 任务实例训练的模仿或强化学习方法, 成功率分别提高了34%和10%。我们还通过展示在仅包含行动的控制基线上始终具有优势, 证明了稀疏、多功能的推理在决策中的重要性。除了通用性和性能提升外, 推理和行动的结还有助于模型的可解释性、可信度和跨所有领域的可诊断性, 因为人类可以轻松区分模型内部知识与外部环境的信息, 并且可以检查推理轨迹以理解模型行动的决策基础。

总结来说, 我们的主要贡献如下: (1) 我们提出了ReAct, 一种基于提示的新范式, 用于在语言模型中协同推理和行动以解决一般任务; (2) 我们在多个基准测试中进行了广泛的实验, 展示了ReAct在少样本学习设置下相对于先前独立执行推理或行动生成方法的优势; (3) 我们提出了系统的消融分析和研究, 以理解行动在推理任务中的重要性, 以及推理在交互任务中的作用; (4) 我们分析了ReAct在提示设置下的局限性（即推理和行动行为的支持有限）, 并进行了初步的微调实验, 展示了ReAct在增加训练数据后改进的潜力。将ReAct扩展到更多任务并与其他互补范式（如强化学习）结合, 可以进一步释放大语言模型的潜力。

2 ReAct: 协同推理与行动 +

考虑一个智能体与环境交互以解决任务的通用设置。在时间步 t , 智能体从环境中接收到观察 $o_t \in \mathcal{O}$ 并根据某种策略 $\pi(a_t|c_t)$ 采取动作 $a_t \in \mathcal{A}$, 其中 $c_t = (o_1, a_1, \dots, o_{t-1}, a_{t-1}, o_t)$ 是智能体的上下文。当映射 $c_t \mapsto a_t$ 高度隐含且需要大量计算时, 学习策略是具有挑战性的。例如, 图1(1c)中显示的智能体无法生成正确的最终动作（动作4）来完成问答任务, 因为它需要对轨迹上下文（问题、动作1-3、观察1-3）进行复杂的推理。类似地, 图1(2a)中显示的智能体未能从上下文中理解到池子1不包含胡椒撒播器1, 因此继续产生幻觉动作。

ReAct 的想法很简单: 我们将智能体的动作空间扩展到 $\hat{\mathcal{A}} = \mathcal{A} \cup \mathcal{L}$, 其中 $\mathcal{L}$ 是语言空间。语言空间中的动作 $\hat{a}_t \in \mathcal{L}$, 我们将之称为思考或推理轨迹, 不会影响外部环境, 因此不会产生观察反馈。相反, 思考 $\hat{a}_t$ 旨在通过推理当前上下文 c_t 来组合有用信息, 并更新上下文 $c_{t+1} = (c_t, \hat{a}_t)$ 以支持未来的推理或行动。如图 1 所示, 可能存在各种类型的有用思考, 例如分解任务目标并创建行动计划 (2b, 行动 1; 1d, 思考 1)、注入与任务解决相关的常识知识 (2b, 行动 1)、从观察中提取重要部分 (1d, 思考 2, 4)、跟踪进度和转换行动计划 (2b, 行动 8)、处理异常并调整行动计划 (1d, 思考 3), 等等。

然而, 由于语言空间 $\mathcal{L}$ 是无限的, 在这个增强动作空间中学习是困难的, 并且需要强大的语言先验知识。在本文中, 我们主要关注这样一种设置: 一个冻结的大型语言模型PaLM-540B (Chowdhery等人, 2022)¹, 通过少量上下文示例提示, 以生成特定领域的动作和用于任务解决的自由形式语言思考 (图1 (1d), (2b))。每个上下文示例是一个人类动作轨迹, 包括动作、思考和环境观察, 用于解决一个任务实例 (见附录C)。对于推理是首要任务的那些任务 (图1(1)), 我们交替生成思考和动作, 以便任务解决轨迹由多个思考-动作-观察步骤组成。相比之下, 对于可能涉及大量动作的决策任务 (图1(2)), 思考只需要出现在轨迹中最相关的位置, 因此我们让语言模型自行决定思考和动作的异步出现。

¹我们在附录A.1中展示了部分GPT-3 (Brown等人, 2020) 的结果, 其表现优于PaLM-540B。

appear sparsely in the most relevant positions of a trajectory, so we let the language model decide the asynchronous occurrence of thoughts and actions for itself.

Since decision making and reasoning capabilities are integrated into a large language model, ReAct enjoys several unique features: **A) Intuitive and easy to design:** Designing ReAct prompts is straightforward as human annotators just type down their thoughts in language on top of their actions taken. No ad-hoc format choice, thought design, or example selection is used in this paper. We detail prompt design for each task in Sections 3 and 4. **B) General and flexible:** Due to the flexible thought space and thought-action occurrence format, ReAct works for diverse tasks with distinct action spaces and reasoning needs, including but not limited to QA, fact verification, text game, and web navigation. **C) Performant and robust:** ReAct shows strong generalization to new task instances while learning solely from one to six in-context examples, consistently outperforming baselines with only reasoning or acting across different domains. We also show in Section 3 additional benefits when finetuning is enabled, and in Section 4 how ReAct performance is robust to prompt selections. **D) Human aligned and controllable:** ReAct promises an interpretable sequential decision making and reasoning process where humans can easily inspect reasoning and factual correctness. Moreover, humans can also control or correct the agent behavior on the go by thought editing, as shown in Figure 5 in Section 4.

3 KNOWLEDGE-INTENSIVE REASONING TASKS

We begin with knowledge-intensive reasoning tasks like multi-hop question answering and fact verification. As shown in Figure 1(1d), by interacting with a Wikipedia API, ReAct is able to retrieve information to support reasoning, while also use reasoning to target what to retrieve next, demonstrating a synergy of reasoning and acting.

3.1 SETUP

Domains We consider two datasets challenging knowledge retrieval and reasoning: (1) Hot-PotQA (Yang et al., 2018), a multi-hop question answering benchmark that requires reasoning over two or more Wikipedia passages, and (2) FEVER (Thorne et al., 2018), a fact verification benchmark where each claim is annotated SUPPORTS, REFUTES, or NOT ENOUGH INFO, based on if there exists a Wikipedia passage to verify the claim. In this work, we operate in a question-only setup for both tasks, where models only receive the question/claim as input without access to support paragraphs, and have to rely on their internal knowledge or retrieve knowledge via interacting with an external environment to support reasoning.

Action Space We design a simple Wikipedia web API with three types of actions to support interactive information retrieval: (1) **search**[entity], which returns the first 5 sentences from the corresponding entity wiki page if it exists, or else suggests top-5 similar entities from the Wikipedia search engine, (2) **lookup**[string], which would return the next sentence in the page containing string, simulating Ctrl+F functionality on the browser. (3) **finish**[answer], which would finish the current task with answer. We note that this action space mostly can only retrieve a small part of a passage based on exact passage name, which is significantly weaker than state-of-the-art lexical or neural retrievers. The purpose is to simulate how humans would interact with Wikipedia, and force models to retrieve via explicit reasoning in language.

3.2 METHODS

ReAct Prompting For HotpotQA and Fever, we randomly select 6 and 3 cases² from the training set and manually compose ReAct-format trajectories to use as few-shot exemplars in the prompts. Similar to Figure 1(d), each trajectory consists of multiple thought-action-observation steps (i.e. dense thought), where free-form thoughts are used for various purposes. Specifically, we use a combination of thoughts that decompose questions (“I need to search x, find y, then find z”), extract information from Wikipedia observations (“x was started in 1844”, “The paragraph does not tell x”), perform commonsense (“x is not y, so z must instead be...”) or arithmetic reasoning (“1844 < 1989”), guide

²We find more examples do not improve performance.

在轨迹中最相关的位置上稀疏地出现，所以我们让语言模型自己决定思想和行动的异步发生。

由于决策能力和推理能力被集成到大型语言模型中，ReAct 具有以下几个独特特点：A) 直观且易于设计：由于人类标注者只需在采取行动的基础上用语言写下他们的想法，因此设计 ReAct 提示非常直接。本文没有使用任何特设的格式选择、思维设计或示例选择。我们在第 3 节和第 4 节详细介绍了每个任务的提示设计。B) 通用且灵活：由于思维空间的灵活性和思维-行动出现格式，ReAct 可用于具有不同行动空间和推理需求的各种任务，包括但不限于问答、事实验证、文本游戏和网络导航。C) 性能优异且鲁棒：ReAct 在仅从一到六个上下文示例中学习的情况下，表现出对新的任务实例的强大泛化能力，始终优于仅使用推理或行动的基线模型，跨不同领域。我们在第 3 节还展示了启用微调时的额外优势，并在第 4 节展示了 ReAct 性能对提示选择的鲁棒性。D) 与人类对齐且可控：ReAct 提供了一个可解释的序列化决策和推理过程，人类可以轻松检查推理和事实的正确性。此外，人类还可以通过思维编辑随时控制或纠正代理行为，如图 5 所示（第 4 节）。

3 知识密集型推理任务

我们从知识密集型推理任务开始，例如多跳问答和事实验证。如图1(1d)所示，通过与维基百科API交互，ReAct能够检索信息以支持推理，同时也能利用推理来决定下一步要检索什么，展示了推理与行动的协同作用。

3.1 设置

领域 我们考虑了两个具有挑战性的知识检索和推理数据集：(1) Hot-PotQA（Yang等人，2018年），一个需要跨两个或多个维基百科段落进行推理的多跳问答基准，以及(2) FEVER（Thorne等人，2018年），一个事实验证基准，其中每个声明根据是否存在维基百科段落来验证声明而被标注为支持、反驳或信息不足。在这项工作中，我们对这两个任务都采用仅处理问题的设置，其中模型仅接收问题/声明作为输入，无法访问支持段落，并且必须依赖其内部知识或通过与环境交互来检索知识以支持推理。

动作空间 我们设计了一个简单的维基百科网页API，包含三种类型的动作来支持交互式信息检索：(1) 搜索[实体]，，如果对应实体的维基百科页面存在，则返回前5句话；否则，会从维基百科搜索引擎建议前5个相似实体，(2) 查找[字符串]，，会返回包含该字符串的页面下一句话，模拟浏览器上的Ctrl+F功能。(3) 完成[答案]，，会以答案完成当前任务。我们注意到这个动作空间主要只能根据精确的段落名称检索一小部分段落，这远弱于当前最先进的词汇或神经检索器。其目的是模拟人类如何与维基百科交互，并迫使模型通过显式语言推理进行检索。

3.2 方法

针对HotpotQA和Fever，我们从训练集中随机选择6个和3个案例²，手动编写ReAct格式的轨迹，用作提示中的少样本示例。类似于图1(d)，每个轨迹由多个思考-行动-观察步骤（即密集思考）组成，自由形式的思考用于各种目的。具体来说，我们使用多种思考的组合，包括分解问题（“我需要搜索x，找到y，然后找到z”）、从维基百科观察中提取信息（“x始于1844年”、“段落没有说明x”）、进行常识推理（“x不是y，所以z必须...”）或算术推理（“1844 < 1989”）、指导搜索重构（“也许我可以搜索/查找x代替”），并综合最终答案（“...所以答案是x”）。更多细节请参见附录C。

²我们发现更多例子并未提升性能。

Prompt Method ^a	HotpotQA (EM)	Fever (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC $\rightarrow$ ReAct	34.2	64.6
ReAct $\rightarrow$ CoT-SC	35.1	62.0
Supervised SoTA^b	67.5	89.5

Table 1: PaLM-540B prompting results on HotpotQA and Fever.

^aHotpotQA EM is 27.1, 28.9, 33.8 for Standard, CoT, CoT-SC in Wang et al. (2022b).

^b(Zhu et al., 2021; Lewis et al., 2020)

search reformulation (“maybe I can search/look up x instead”), and synthesize the final answer (“...so the answer is x”). See Appendix C for more details.

Baselines We systematically ablate ReAct trajectories to build prompts for multiple baselines (with formats as Figure 1(1a-1c)): (a) **Standard prompting** (Standard), which removes all thoughts, actions, observations in ReAct trajectories. (b) **Chain-of-thought prompting** (CoT) (Wei et al., 2022), which removes actions and observations and serve as a reasoning-only baseline. We also build a self-consistency baseline (CoT-SC) (Wang et al., 2022a;b) by sampling 21 CoT trajectories with decoding temperature 0.7 during inference and adopting the majority answer, which is found to consistently boost performance over CoT. (c) **Acting-only prompt** (Act), which removes thoughts in ReAct trajectories, loosely resembling how WebGPT (Nakano et al., 2021) interacts with the Internet to answer questions, though it operates on a different task and action space, and uses imitation and reinforcement learning instead of prompting.

Combining Internal and External Knowledge As will be detail in Section 3.3, we observe that the problem solving process demonstrated by ReAct is more factual and grounded, whereas CoT is more accurate in formulating reasoning structure but can easily suffer from hallucinated facts or thoughts. We therefore propose to incorporate ReAct and CoT-SC, and let the model decide when to switch to the other method based on the following heuristics: A) **ReAct $\rightarrow$ CoT-SC**: when ReAct fails to return an answer within given steps, back off to CoT-SC. We set 7 and 5 steps for HotpotQA and FEVER respectively as we find more steps will not improve ReAct performance³. B) **CoT-SC $\rightarrow$ ReAct**: when the majority answer among n CoT-SC samples occurs less than $n/2$ times (i.e. internal knowledge might not support the task confidently), back off to ReAct.

Finetuning Due to the challenge of manually annotating reasoning traces and actions at scale, we consider a bootstrapping approach similar to Zelikman et al. (2022), using 3,000 trajectories with correct answers generated by ReAct (also for other baselines) to finetune smaller language models (PaLM-8/62B) to decode trajectories (all thoughts, actions, observations) conditioned on input questions/claims. More details are in Appendix B.1.

3.3 RESULTS AND OBSERVATIONS

ReAct outperforms Act consistently Table 1 shows HotpotQA and Fever results using PaLM-540B as the base model with different prompting methods. We note that ReAct is better than Act on both tasks, demonstrating the value of reasoning to guide acting, especially for synthesizing the final answer, as shown in Figure 1 (1c-d). Fine-tuning results 3 also confirm the benefit of reasoning traces for more informed acting.

³Of all trajectories with correct final answers, those with 7 steps on HotpotQA and 5 steps on FEVER only take up 0.84% and 1.33% respectively.

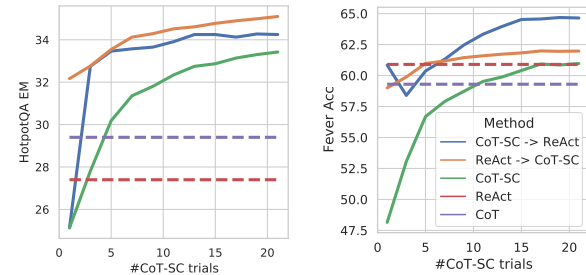


Figure 2: PaLM-540B prompting results with respect to number of CoT-SC samples used.

提示方法 ^a	HotpotQA (EM)	Fever (Acc)
标准	28.7	57.1
CoT (Wei 等, 2022)	29.4	56.3
CoT-SC (王等, 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC $\rightarrow$ ReAct	34.2	64.6
ReAct $\rightarrow$ CoT-SC	35.1	62.0
监督式当前最佳^b	67.5	89.5

表1: PaLM-540B在HotpotQA和Fever上的提示结果。

^aHotpotQA EM is 27.1, 28.9, 33.8 for Standard, CoT, CoT-SC in Wang et al. (2022b).

^b(Zhu et al., 2021; Lewis et al., 2020)

搜索重构 (“也许我可以搜索/查找x代替”), 并综合最终答案 (“...所以答案是x”)。更多细节请参见附录C。

基线 我们系统地消融ReAct轨迹来构建多个基线的提示 (格式如图1(1a-1c)所示): (a) 标准提示 (Standard), 移除ReAct轨迹中的所有思考、行动、观察。(b) 思维链提示 (CoT) (Wei等人, 2022), 移除行动和观察, 作为仅推理的基线。我们还构建了一个自治基线 (CoT-SC) (Wang等人, 2022a;b), 通过在推理时采样21个CoT轨迹, 解码温度为0.7, 并采用多数答案, 发现这始终能提升性能超过CoT。(c) 仅行动提示 (Act), 移除ReAct轨迹中的思考, 大致类似于WebGPT (Nakano等人, 2021) 如何与互联网交互来回答问题, 尽管它作用于不同的任务和行动空间, 并使用模仿和强化学习而非提示。

结合内部和外部知识 如第3.3节将详细说明, 我们观察到ReAct展示的问题解决过程更注重事实和基础, 而CoT在构建推理结构方面更准确, 但容易产生幻觉的事实或想法。因此, 我们提出结合ReAct和CoT-SC, 并让模型根据以下启发式规则决定何时切换到另一种方法: A) **ReAct $\rightarrow$ CoT-SC**: 当ReAct在规定步骤内无法返回答案时, 退回到CoT-SC。我们为HotpotQA和FEVER分别设置了7和5步, 因为我们发现增加步骤不会提升ReAct的性能³。B) **CoT-SC $\rightarrow$ ReAct**: 当 n CoT-SC样本中的多数答案出现次数少于 $n/2$ 次 (即内部知识可能无法自信地支持任务) 时, 退回到ReAct。

微调 由于手动标注推理轨迹和动作在大规模上的挑战, 我们考虑了一种类似于Zelikman等人 (2022) 的引导式方法, 使用ReAct (以及其他基线) 生成的3,000个带正确答案的轨迹来微调较小的语言模型 (PaLM-8/62B), 以解码基于输入问题/主张的条件轨迹 (所有思考、动作、观察)。更多细节见附录B.1。

3.3 结果与观察

ReAct 持续优于 Act 表1展示了使用 PaLM-540B 作为基础模型并结合不同提示方法在 HotpotQA 和 Fever 上的结果。我们注意到 ReAct 在两项任务上都优于 Act, 这证明了推理对行动的指导价值, 尤其是在综合最终答案方面, 如图1 (1c-d) 所示。微调结果3也证实了推理轨迹对更明智行动的益处。

³在所有最终答案正确的轨迹中, HotpotQA上7步、FEVER上5步的轨迹分别仅占0.84%和1.33%。

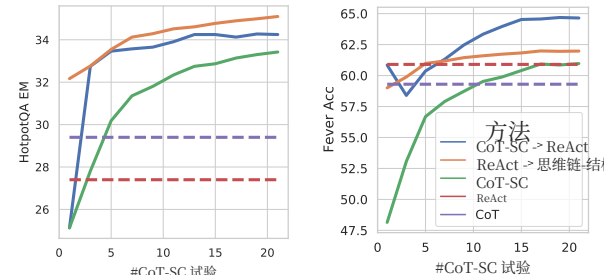


图2: PaLM-540B提示结果与所用CoT-SC样本数量的关系。

	Type	Definition	ReAct	CoT
Success	True positive	Correct reasoning trace and facts	94%	86%
	False positive	Hallucinated reasoning trace or facts	6%	14%
Failure	Reasoning error	Wrong reasoning trace (including failing to recover from repetitive steps)	47%	16%
	Search result error	Search return empty or does not contain useful information	23%	-
	Hallucination	Hallucinated reasoning trace or facts	0%	56%
	Label ambiguity	Right prediction but did not match the label precisely	29%	28%

Table 2: Types of success and failure modes of ReAct and CoT on HotpotQA, as well as their percentages in randomly selected examples studied by human.

ReAct vs. CoT On the other hand, ReAct outperforms CoT on Fever (60.9 vs. 56.3) and slightly lags behind CoT on HotpotQA (27.4 vs. 29.4). Fever claims for SUPPORTS/REFUTES might only differ by a slight amount (see Appendix D.1), so acting to retrieve accurate and up-to-date knowledge is vital. To better understand the behavioral difference between ReAct and CoT on HotpotQA, we randomly sampled 50 trajectories with correct and incorrect answers (judged by EM) from ReAct and CoT respectively (thus 200 examples in total), and manually labeled their success and failure modes in Table 2. Some key observations are as follows:

A) **Hallucination is a serious problem for CoT**, resulting in much higher false positive rate than ReAct (14% vs. 6%) in success mode, and make up its major failure mode (56%). In contrast, the problem solving trajectory of ReAct is more grounded, fact-driven, and trustworthy, thanks to the access of an external knowledge base.

B) **While interleaving reasoning, action and observation steps improves ReAct’s groundedness and trustworthiness, such a structural constraint also reduces its flexibility in formulating reasoning steps**, leading to more reasoning error rate than CoT. we note that there is one frequent error pattern specific to ReAct, in which the model repetitively generates the previous thoughts and actions, and we categorize it as part of “reasoning error” as the model fails to reason about what the proper next action to take and jump out of the loop⁴.

C) **For ReAct, successfully retrieving informative knowledge via search is critical**. Non-informative search, which counts for 23% of the error cases, derails the model reasoning and gives it a hard time to recover and reformulate thoughts. This is perhaps an expected trade-off between factuality and flexibility, which motivates our proposed strategies of combining two methods.

We provide examples for each success and failure modes in Appendix E.1. We also find some HotpotQA questions may contain outdated answer labels, see Figure 4 for example.

ReAct + CoT-SC perform best for prompting LLMs Also shown in Table 1, the best prompting method on HotpotQA and Fever are ReAct → CoT-SC and CoT-SC → ReAct respectively. Furthermore, Figure 2 shows how different methods perform with respect to the number of CoT-SC samples used. While two ReAct + CoT-SC methods are advantageous at one task each, they both significantly and consistently outperform CoT-SC across different number of samples, reaching CoT-SC performance with 21 samples using merely 3-5 samples. These results indicate the value of properly combining model internal knowledge and external knowledge for reasoning tasks.

ReAct performs best for fine-tuning Figure 3 shows the scaling effect of prompting/finetuning four methods (Standard, CoT, Act, ReAct) on HotpotQA. With PaLM-8/62B, prompting ReAct performs worst among four methods due to the difficulty to learn both reasoning and acting from in-context examples. However, when finetuned with just 3,000 examples, ReAct becomes the best method among the four, with PaLM-8B finetuned ReAct outperforming all PaLM-62B prompting methods, and PaLM-62B finetuned ReAct outperforming all 540B prompting methods. In contrast, finetuning Standard or CoT is significantly worse than finetuning ReAct or Act for both PaLM-8/62B, as the former essentially teaches models to memorize (potentially hallucinated) knowledge facts, and the latter teaches models how to (reason and) act to access information from Wikipedia, a more generalizable skill for knowledge reasoning. As all prompting methods are still significantly far from domain-specific state-of-the-art approaches (Table 1), we believe finetuning with more human-written data might be a better way to unleash the power of ReAct.

⁴We suspect that this could be due to the sub-optimal greedy decoding procedure, and future work using better decoding (e.g. beam search) might help address this issue.

	Type	定义	ReAct	CoT
成功	真阳性	正确的推理轨迹和事实	94%	86%
	假阳性	幻觉推理轨迹或事实	6%	14%
失败	推理错误	错误的推理轨迹（包括无法从重复步骤中恢复）	47%	16%
	搜索结果错误	搜索结果为空或不包含有用信息	23%	-
	幻觉	幻觉推理轨迹或事实	0%	56%
	标签歧义	预测正确但未精确匹配标签	29%	28%

表2: ReAct和CoT在HotpotQA上的成功和失败模式类型，以及它们在人类随机研究样本中的百分比。

ReAct 与 CoT 另一方面，ReAct 在 Fever 上优于 CoT（60.9 vs. 56.3），但在 HotpotQA 上略逊于 CoT（27.4 vs. 29.4）。Fever 中 SUPPORTS/REFUTES 的主张可能仅相差微小（参见附录 D.1），因此检索准确且最新的知识至关重要。为了更好地理解 ReAct 和 CoT 在 HotpotQA 上的行为差异，我们从 ReAct 和 CoT 中分别随机采样了50个正确和错误的轨迹（由EM判断）（总共200个示例），并在表2中手动标注了它们的成功和失败模式。一些关键观察如下：

A) 幻觉是 CoT 的一个严重问题，导致其在成功模式下的假阳性率远高于 ReAct（14 % vs. 6%），并构成其主要失效模式（56%）。相比之下，ReAct 的问题解决轨迹更加基于现实、以事实为驱动且值得信赖，这得益于其访问外部知识库的能力。

B) 虽然交错推理、行动和观察步骤可以提高 ReAct 的根基性和可信度，但这种结构约束也降低了它在制定推理步骤时的灵活性，导致其推理错误率高于 CoT。我们注意到 ReAct 存在一个特定的频繁错误模式，即模型反复生成之前的想法和行动，我们将其归类为“推理错误”的一部分，因为模型未能推理出应采取的正确下一步行动并跳出循环⁴。

C) 对于ReAct，通过搜索成功获取信息性知识至关重要。非信息性搜索占错误案例的23%，会中断模型的推理过程，并使其难以恢复和重新组织思路。这或许是在f之间的一种预期权衡。

实际类型d flexibility, which motivates我们的提议 d 策略 ies of combining two methods. 我们在附录E.1中为每种成功和失败模式提供了示例。我们还发现一些HotpotQA问题可能包含过时的答案标签，例如，参见图4。

ReAct + CoT-SC在提示LLM方面表现最佳。如表1所示，在HotpotQA和Fever上表现最佳的提示方法是ReAct → CoT-SC和CoT-SC → ReAct。此外，图2展示了不同方法在使用不同数量的CoT-SC样本时的表现。虽然两种ReAct + CoT-SC方法在各自的任务上有优势，但它们在不同数量的样本下都显著且持续地优于CoT-SC，仅使用3-5个样本就达到了使用21个样本的CoT-SC性能。这些结果表明，在推理任务中，正确结合模型内部知识和外部知识具有重要价值。

ReAct 在微调方面表现最佳。图 3 展示了提示/微调四种方法（标准、CoT、Act、ReAct）在 HotpotQA 上的缩放效应。使用 PaLM-8/62B 时，提示 ReAct 在四种方法中表现最差，因为从上下文示例中学习推理和行动都存在困难。然而，当仅使用 3,000 个示例进行微调时，ReAct 成为四种方法中表现最好的方法，其中微调后的 PaLM-8B ReAct 表现优于所有 PaLM-62B 提示方法，而微调后的 PaLM-62B ReAct 表现优于所有 540B 提示方法。相比之下，对于 PaLM-8/62B，微调标准或 CoT 的效果明显不如微调 ReAct 或 Act，因为前者本质上教模型记忆（可能存在幻觉的）知识事实，而后者教模型如何（推理和）行动以从维基百科获取信息，这是一种更通用的知识推理技能。由于所有提示方法与特定领域的最先进方法仍有显著差距（表 1），我们认为使用更多人工编写的数据进行微调可能是释放 ReAct 力量的更好方法。

⁴我们怀疑这可能是由于次优的贪婪解码过程，未来使用更好的解码方法（例如，束搜索）可能有助于解决这个问题。

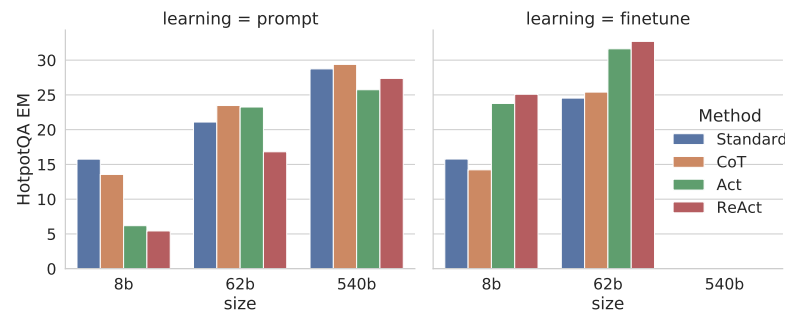


Figure 3: Scaling results for prompting and finetuning on HotPotQA with ReAct (ours) and baselines.

4 DECISION MAKING TASKS

We also test ReAct on two language-based interactive decision-making tasks, ALFWorld and WebShop, both of which feature complex environments that require agents to act over long horizons with sparse rewards, warranting the need for reasoning to act and explore effectively.

ALFWorld ALFWorld (Shridhar et al., 2020b) (Figure 1(2)) is a synthetic text-based game designed to align with the embodied ALFRED benchmark (Shridhar et al., 2020a). It includes 6 types of tasks in which an agent needs to achieve a high-level goal (e.g. examine paper under desk lamp) by navigating and interacting with a simulated household via text actions (e.g. go to coffeetable 1, take paper 2, use desk lamp 1). A task instance can have more than 50 locations and take an expert policy more than 50 steps to solve, thus challenging an agent to plan and track subgoals, as well as explore systematically (e.g. check all desks one by one for desk lamp). In particular, one challenge built into ALFWorld is the need to determine likely locations for common household items (e.g. desk lamps will likely be on desks, shelves, or dressers), making this environment a good fit for LLMs to exploit their pretrained commonsense knowledge. To prompt ReAct, we randomly annotate three trajectories from the training set for each task type, where each trajectory includes sparse thoughts that (1) decompose the goal, (2) track subgoal completion, (3) determine the next subgoal, and (4) reason via commonsense where to find an object and what to do with it. We show prompts used for ALFWorld in Appendix C.4. Following Shridhar et al. (2020b), we evaluate on 134 unseen evaluation games in a task-specific setup. For robustness, we construct 6 prompts for each task type through each permutation of 2 annotated trajectories from the 3 we annotate. Act prompts are constructed using the same trajectories, but without thoughts — since task instances are randomly chosen from the training set, it favors neither ReAct nor Act and provides a fair and controlled comparison to test the importance of sparse thoughts. For baselines, we use BUTLER (Shridhar et al., 2020b), an imitation learning agent trained on 10^5 expert trajectories for each task type⁵.

WebShop Can ReAct also interact with noisy real-world language environments for practical applications? We investigate WebShop (Yao et al., 2022), a recently proposed online shopping website environment with 1.18M real-world products and 12k human instructions. Unlike ALFWorld, Webshop contains a high variety of structured and unstructured texts (e.g. product titles, descriptions, and options crawled from Amazon), and requires an agent to purchase a product based on a user instruction (e.g. “I am looking for a nightstand with drawers. It should have a nickel finish, and priced lower than \$140”) through web interactions (e.g. search “nightstand drawers”, choose buttons such as “color: modern-nickel-white” or “back to search”). This task is evaluated by average score (percentage of desired attributes covered by the chosen product averaged across all episodes) and success rate (percentage of episodes where the chosen product satisfies all requirements) on 500 test instructions. We formulate Act prompts with actions to search, choose product, choose options, and buy, with ReAct prompts additionally reasoning to determine what to explore, when to buy, and what products options are relevant to the instruction. See Table 6 for an example prompt, and Table 10 for model predictions in the Appendix. We compare to an imitation learning (IL) method

⁵Micheli & Fleuret (2021) finetuned a GPT-2 model on 3553 task instances and achieved a much improved performance than BUTLER, but it is trained on all task types, thus not included as a baseline.

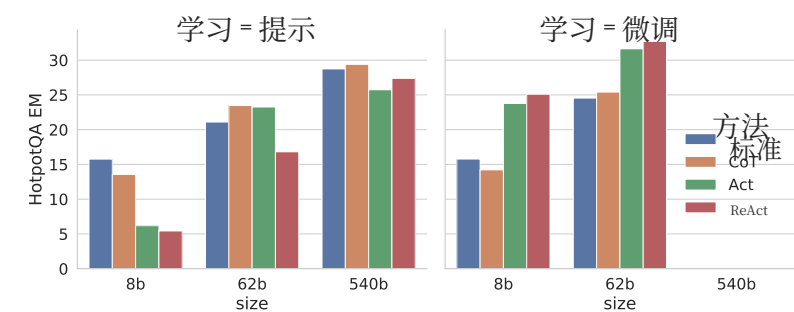


图3：在HotPotQA上使用ReAct（ours）和基线进行提示和微调的扩展结果。

4 决策任务

我们还测试了 ReAct 在两个基于语言的交互式决策任务 ALFWorld 和 WebShop 上的表现，这两个任务都涉及复杂环境，需要代理在长时程内以稀疏奖励行动，这需要推理来有效行动和探索。

ALFWorld ALFWorld (Shridhar等人, 2020b) (图1(2))是一个合成文本游戏，旨在与具身 ALFRED基准 (Shridhar等人, 2020a) 保持一致。它包含6种类型的任务，其中代理需要通过文本动作（例如：去到1号咖啡桌，拿起2号纸，使用1号台灯）导航并与模拟家庭环境交互，以实现高级目标（例如：在台灯下检查纸）。一个任务实例可以包含超过50个位置，需要专家策略超过50步才能解决，因此挑战代理进行子目标规划和跟踪，以及系统性地探索（例如：逐一检查所有桌子是否有台灯）。特别是，ALFWorld中构建的一个挑战是需要确定常见家用物品的可能位置（例如：台灯可能位于桌子、架子或衣柜上），这使得该环境成为LLM利用其预训练常识知识的良好选择。为了提示 ReAct，我们对每种任务类型随机标注三个训练集中的轨迹，其中每个轨迹包含稀疏思考，这些思考（1）分解目标，（2）跟踪子目标完成情况，（3）确定下一个子目标，以及（4）通过常识推理确定在哪里找到对象以及如何处理它。我们在附录C.4中展示了用于ALFWorld的提示。遵循Shridhar等人（2020b），我们在134个未见过的评估游戏中以特定任务设置进行评估。为了稳健性，我们通过对比标注的3个轨迹进行所有排列组合，为每种任务类型构建6个提示。Act提示使用相同的轨迹构建，但没有思考——由于任务实例是从训练集中随机选择的，它既不偏向ReAct也不偏向Act，提供了公平和受控的比较，以测试稀疏思考的重要性。对于基线，我们使用BUTLER (Shridhar等人, 2020b)，这是一个模仿学习代理，它在每个任务类型上使用 10^5 专家轨迹⁵进行训练。

WebShop Can ReAct 还能与嘈杂的真实世界语言环境交互，用于实际应用吗？我们研究了 WebShop (Yao 等人, 2022 年)，这是一个最近提出的在线购物网站环境，包含 118 万个真实世界的产品和 12 千条人类指令。与 ALFWorld 不同，Webshop 包含多种结构化和非结构化文本（例如从亚马逊抓取的产品标题、描述和选项），并要求代理根据用户指令（例如“我正在寻找一个带抽屉的床头柜。它应该有镍色抛光，价格低于 140 美元”）通过网页交互（例如搜索“床头柜抽屉”，选择按钮如“颜色：现代镍白色”或“返回搜索”）来购买产品。该任务通过在 500 条测试指令上的平均分数（所选产品满足期望属性的比例）和成功率（所选产品满足所有要求的指令比例）进行评估。我们制定了包含搜索、选择产品、选择选项和购买等操作的 Act 提示，ReAct 提示则额外推理以确定要探索的内容、何时购买以及哪些产品选项与指令相关。参见表 6 的示例提示，以及附录中的表 10 的模型预测。我们将其与一个模仿学习（IL）方法进行比较

⁵米歇利和弗勒雷特（2021）在3553个任务实例上微调了一个GPT-2模型，性能比BUTLER提升了很多，但它训练了所有任务类型，因此不作为基线包含在内。

Method	Pick	Clean	Heat	Cool	Look	Pick 2	All
Act (best of 6)	88	42	74	67	72	41	45
ReAct (avg)	65	39	83	76	55	24	57
ReAct (best of 6)	92	58	96	86	78	41	71
ReAct-IM (avg)	55	59	60	55	23	24	48
ReAct-IM (best of 6)	62	68	87	57	39	33	53
BUTLER _g (best of 8)	33	26	70	76	17	12	22
BUTLER (best of 8)	46	39	74	100	22	24	37

Table 3: AlfWorld task-specific success rates (%). BUTLER and BUTLER_g results are from Table 4 of Shridhar et al. (2020b). All methods use greedy decoding, except that BUTLER uses beam search.

Method	Score	SR
Act	62.3	30.1
ReAct	66.6	40.0
IL	59.9	29.1
IL+RL	62.4	28.7
Human Expert	82.1	59.6

Table 4: Score and success rate (SR) on Webshop. IL/IL+RL taken from Yao et al. (2022).

trained with 1,012 human annotated trajectories, and a imitation + reinforcement learning (IL + RL) method additionally trained with 10,587 training instructions.

Results ReAct outperforms Act on both ALFWorld (Table 3) and Webshop (Table 4). On ALFWorld, the best ReAct trial achieves an average success rate of 71%, significantly outperforming the best Act (45%) and BUTLER (37%) trials. In fact, even the worse ReAct trial (48%) beats the best trial of both methods. Moreover, the advantage of ReAct over Act is consistent across six controlled trials, with relative performance gain ranging from 33% to 90% and averaging 62%. Qualitatively, we saw that, without any thoughts at all, Act fails to correctly decompose goals into smaller subgoals, or loses track of the current state of the environment. Example trajectories comparing ReAct and Act can be found in Appendix D.2.1 and Appendix D.2.2.

On Webshop, one-shot Act prompting already performs on par with IL and IL+RL methods. With additional sparse reasoning, ReAct achieves significantly better performance, with an absolute 10% improvement over the previous best success rate. By checking examples, we find that ReAct is more likely to identify instruction-relevant products and options by reasoning to bridge the gap between noisy observations and actions (e.g. “For ‘space-saving ottoman bench for living room’, the item has options ‘39x18x18inch’ and ‘blue’ and seems good to buy.”). However, existing methods are still far from the performance of expert humans (Table 4), who perform significantly more product explorations and query re-formulations that are still challenging for prompting-based methods.

On the value of internal reasoning vs. external feedback To our knowledge, ReAct is the first demonstration of combined reasoning and action using an LLM applied to an interactive environment within a closed-loop system. Perhaps the closest prior work is Inner Monologue (IM), from Huang et al. (2022b), in which actions from an embodied agent are motivated by an eponymous “inner monologue”. **However, IM’s “inner monologue” is limited to observations of the environment state and what needs to be completed by the agent for the goal to be satisfied.** In contrast, the reasoning traces in ReAct for decision making is flexible and sparse, allowing diverse reasoning types (see Section 2) to be induced for different tasks.

To demonstrate the differences between ReAct and IM, and to highlight the importance of internal reasoning vs. simple reactions to external feedback, we ran an ablation experiment using a thought pattern composed of IM-like dense external feedback. As can be seen in Table 3, ReAct substantially outperforms IM-style prompting (ReAct-IM) (71 vs. 53 overall success rate), with consistent advantages on five out of six tasks. Qualitatively, we observed that ReAct-IM often made mistakes in identifying when subgoals were finished, or what the next subgoal should be, due to a lack of high-level goal decomposition. Additionally, many ReAct-IM trajectories struggled to determine where an item would likely be within the ALFWorld environment, due to a lack of commonsense reasoning. Both shortcomings can be addressed in the ReAct paradigm. More details about ReAct-IM is in Appendix B.2. An example prompt for ReAct-IM can be found in Appendix C.4, and an example trajectory in Appendix D.2.3.

方法	Pick	干净	Heat	Cool	Look	选择2	All
行动（最佳6局）	88	42	74	67	72	41	45
再行动（平均）	65	39	83	76	55	24	57
ReAct（前6名最佳）	92	58	96	86	78	41	71
ReAct-IM（平均）	55	59	60	55	23	24	48
ReAct-IM（前6名最佳）	62	68	87	57	39	33	53
BUTLER _g （前8名最佳）	33	26	70	76	17	12	22
BUTLER（前8名最佳）	46	39	74	100	22	24	37

表3: AlfWorld任务特定成功率(%). BUTLER和BUTLER_g的结果来自Shridhar等人(2020b)的表4。所有方法均使用贪婪解码，但BUTLER使用束搜索。

方法	得分	SR
Act	62.3	30.1
ReAct	66.6	40.0
IL	59.9	29.1
IL+RL	62.4	28.7
人类专家	82.1	59.6

表4: Webshop上的得分和成功率（SR）。IL/IL+RL数据来自Yao等人（2022年）。

训练了 1,012 条人类标注轨迹，以及一个额外的 aimitation + 强化学习（IL + RL）方法，该方法额外训练了 10,587 条训练指令。

结果表明ReAct在ALFWorld（表3）和Webshop（表4）上均优于Act。在ALFWorld上，ReAct的最佳试验实现了71%的平均成功率，显著优于Act的最佳试验（45%）和BUTLER（37%）试验。事实上，即使ReAct最差的试验（48%）也优于两种方法的最佳试验。此外，ReAct相对于Act的优势在六个受控试验中保持一致，相对性能提升范围在33%至90%，平均为62%。从定性角度来看，我们发现Act在没有任何思考的情况下，无法正确地将目标分解为更小的子目标，或者会丢失环境当前状态的信息。ReAct和Act的比较示例轨迹可以在附录D.2.1和附录D.2.2中找到。

在Webshop上，单次行动提示的性能已经与IL和IL+RL方法相当。通过额外的稀疏推理，ReAct实现了显著更好的性能，相较于之前的最佳成功率提升了10%。通过检查示例，我们发现ReAct更倾向于通过推理来识别与指令相关的产品选项，以弥合噪声观察和行动之间的差距（例如：“对于‘节省空间的客厅长椅’，该商品有‘39x18x18英寸’和‘蓝色’的选项，看起来很适合购买。”）。然而，现有方法仍远不及专家人类的性能（见表4），后者进行显著更多的产品探索和查询重写，而这些对于基于提示的方法来说仍然具有挑战性。

关于内部推理与外部反馈的价值 据我们所知，ReAct是首个在闭环系统中将大型语言模型（LLM）应用于交互式环境，以实现推理和行动结合的演示。或许最接近的前人工作是Huang等人（2022b）提出的Inner Monologue（IM），其中具身代理的行动是由同名的“内心独白”驱动的。然而，IM的“内心独白”仅限于环境状态观察以及代理为满足目标需要完成的事项。相比之下，ReAct用于决策的推理轨迹是灵活且稀疏的，允许针对不同任务诱导多样化的推理类型（见第2节）。

为了展示 ReAct 与 IM 之间的差异，并强调内部推理与对外部反馈的简单反应的重要性，我们进行了一项消融实验，使用了由 IM 类似密集外部反馈组成的思维模式。如表 3 所示，ReAct 的整体成功率（71 vs. 53）显著优于 IM 风格的提示（ReAct-IM），并在六项任务中的五项上表现出持续优势。从定性角度来看，我们发现 ReAct-IM 在识别子目标何时完成或下一个子目标应该是什么时经常出错，这是由于缺乏高层目标分解。此外，许多 ReAct-IM 轨迹难以确定在 ALFWorld 环境中一个物品可能的位置，这是由于缺乏常识推理。这两个缺点都可以在 ReAct 范式中得到解决。有关 ReAct-IM 的更多细节在附录 B.2 中。ReAct-IM 的一个示例提示可以在附录 C.4 中找到，一个示例轨迹在附录 D.2.3 中。

5 RELATED WORK

Language model for reasoning Perhaps the most well-known work of using LLMs for reasoning is Chain-of-Thought (CoT) (Wei et al., 2022), which reveals the ability of LLMs to formulate their own “thinking procedure” for problem solving. Several follow-up works have since been performed, including least-to-most prompting for solving complicated tasks (Zhou et al., 2022), zero-shot-CoT (Kojima et al., 2022), and reasoning with self-consistency (Wang et al., 2022a). Recently, (Madaan & Yazdanbakhsh, 2022) systematically studied the formulation and structure of CoT, and observed that the presence of symbols, patterns and texts is crucial to the effectiveness of CoT. Other work has also been extended to more sophisticated reasoning architecture beyond simple prompting. For example Selection-Inference (Creswell et al., 2022) divides the reasoning process into two steps of “selection” and “inference”. STaR (Zelikman et al., 2022) bootstraps the reasoning process by finetuning the model on correct rationales generated by the model itself. Faithful reasoning (Creswell & Shanahan, 2022) decomposes multi-step reasoning into three steps, each performed by a dedicated LM respectively. Similar approaches like Scratchpad (Nye et al., 2021), which finetunes a LM on intermediate computation steps, also demonstrate improvement on multi-step computation problems. In contrast to these methods, ReAct performs more than just isolated, fixed reasoning, and integrates model actions and their corresponding observations into a coherent stream of inputs for the model to reason more accurately and tackle tasks beyond reasoning (e.g. interactive decision making).

Language model for decision making The strong capability of LLMs has enabled them to perform tasks beyond language generation, and it is becoming more popular to take advantage of LLMs as a policy model for decision making, especially in interactive environments. WebGPT (Nakano et al., 2021) uses an LM to interact with web browsers, navigate through web pages, and infer answers to complicated questions from ELI5 (Fan et al., 2019). In comparison to ReAct, WebGPT does not explicitly model the thinking and reasoning procedure, instead rely on expensive human feedback for reinforcement learning. In conversation modeling, chatbots like BlenderBot (Shuster et al., 2022b) and Sparrow (Glaese et al., 2022) and task-oriented dialogue systems like SimpleTOD (Hosseini-Asl et al., 2020) also train LMs to make decision about API calls. Unlike ReAct, they do not explicitly consider the reasoning procedure either, and also relies on expensive datasets and human feedback collections for policy learning. In contrast, ReAct learns a policy in a much cheaper way, since the decision making process only requires language description of the reasoning procedure.⁶

LLMs have also been increasingly employed in interactive and embodied environments for planning and decision making. Perhaps most relevant to ReAct in this respect are SayCan (Ahn et al., 2022) and Inner Monologue (Huang et al., 2022b), which use LLMs for robotic action planning and decision making. In SayCan, LLMs were prompted to directly predict possible actions a robot can take, which is then reranked by an affordance model grounded on the visual environments for final prediction. Inner Monologue made further improvements by adding the eponymous “inner monologue”, which is implemented as injected feedback from the environment. To our knowledge, Inner Monologue is the first work that demonstrates such a closed-loop system, which ReAct builds on. However, we argue that Inner Monologue does not truly comprise of inner thoughts — this is elaborated in Section 4. We also note that leveraging language as semantically-rich inputs in the process of interactive decision making has been shown to be successful under other settings (Abramson et al., 2020; Karamcheti et al., 2021; Huang et al., 2022a; Li et al., 2022). It is becoming more evident that with the help of LLMs, language as a fundamental cognitive mechanism will play a critical role in interaction and decision making. What is more, progress in LLMs has also inspired the development of versatile and generalist agents like Reed et al. (2022).

6 CONCLUSION

We have proposed ReAct — a simple yet effective method for synergizing reasoning and acting in large language models. Through a diverse set of experiments on multi-hop question-answering, fact checking, and interactive decision-making tasks, we show that ReAct leads to superior performance with interpretable decision traces. Despite the simplicity of our method, complex tasks with large action spaces require more demonstrations to learn well, which unfortunately can easily go beyond the input length limit of in-context learning. We explore the fine-tuning approach on HotpotQA

⁶Human feedback can also be incorporated in a complementary manner but we leave it for future work.

5 相关工作

用于推理的语言模型 也许最著名的使用 LLM 进行推理的工作是思维链 (CoT) (Wei 等人, 2022), 它揭示了 LLM 在为问题解决制定自己的“思维过程”方面的能力。此后, 已经进行了几项后续工作, 包括用于解决复杂任务的从最少到最多提示 (Zhou 等人, 2022)、零样本 CoT (Kojima 等人, 2022) 以及使用自洽性进行推理 (Wang 等人, 2022a)。最近, (Madaan & Yazdanbakhsh, 2022) 系统地研究了 CoT 的制定和结构, 并观察到符号、模式和文本的存在对 CoT 的有效性至关重要。其他工作也已扩展到比简单提示更复杂的推理架构。例如, 选择-推理 (Creswell 等人, 2022) 将推理过程分为“选择”和“推理”两个步骤。STaR (Zelikman 等人, 2022) 通过在模型自身生成的正确推理上微调模型来启动推理过程。忠实推理 (Creswell & Shanahan, 2022) 将多步推理分解为三个步骤, 每一步分别由一个专用的 LM 执行。类似的 Scratchpad (Nye 等人, 2021) 方法, 通过在中间计算步骤上微调 LLM, 也在多步计算问题上表现出改进。与这些方法相比, ReAct 不仅仅是进行孤立的、固定的推理, 而是将模型动作及其对应的观察结果整合为一个连贯的输入流, 使模型能够更准确地推理并处理超出推理范围的任务 (例如, 交互式决策)。

用于决策的语言模型 语言模型 (LLM) 强大的能力使其能够执行超越语言生成任务的操作, 利用语言模型作为决策策略模型正变得越来越普遍, 特别是在交互式环境中。WebGPT (Nakano 等人, 2021 年) 使用语言模型与网络浏览器交互, 浏览网页, 并从 ELI5 (Fan 等人, 2019 年) 中推断复杂问题的答案。与 ReAct 相比, WebGPT 没有显式地建模思考和推理过程, 而是依赖昂贵的反馈进行强化学习。在对话建模中, 像 BlenderBot (Shuster 等人, 2022b) 和 Sparrow (Glaese 等人, 2022 年) 这样的聊天机器人和像 SimpleTOD (Hosseini-Asl 等人, 2020 年) 这样的任务导向对话系统也训练语言模型来决定 API 调用。与 ReAct 不同, 它们也没有显式地考虑推理过程, 并且也依赖昂贵的数集和人工反馈收集进行策略学习。相比之下, ReAct 以更便宜的方式学习策略, 因为决策过程只需要推理过程的语言描述。⁶

大语言模型 (LLM) 也越来越多地应用于交互式 and 具身环境中, 用于规划和决策。在这方面, 与 ReAct 最相关的可能是 SayCan (Ahn 等人, 2022) 和 Inner Monologue (Huang 等人, 2022b), 它们使用 LLM 进行机器人动作规划和决策。在 SayCan 中, LLM 被提示直接预测机器人可以采取的可能动作, 然后由基于视觉环境的可供性模型重新排序以进行最终预测。Inner Monologue 通过添加同名的“内心独白”进行了进一步改进, 该“内心独白”作为来自环境的注入反馈实现。据我们所知, Inner Monologue 是第一个展示此类闭环系统的作品, ReAct 正是基于该系统构建的。然而, 我们认为 Inner Monologue 并不真正包含内心想法——这将在第 4 节中详细阐述。我们还注意到, 在交互式决策过程中将语言作为语义丰富的输入已被证明在其他设置下是成功的 (Abramson 等人, 2020; Karamcheti 等人, 2021; Huang 等人, 2022a; Li 等人, 2022)。随着 LLM 的帮助, 语言作为一种基本认知机制, 在交互和决策中将发挥关键作用。此外, LLM 的进展也启发了通用型智能体 (如 Reed 等人, 2022) 的开发。

6 结论

我们提出了 ReAct——一种简单但有效的协同推理和行动方法, 用于大型语言模型。通过在多跳问答、事实核查和交互式决策任务上的一组多样化实验, 我们展示了 ReAct 能够实现更优的性能, 并具有可解释的决策轨迹。尽管我们的方法很简单, 但具有大动作空间的复杂任务需要更多示例才能良好学习, 不幸的是, 这很容易超出情境学习的输入长度限制。我们探索了在 HotpotQA 上的微调方法

⁶人类反馈也可以以补充的方式进行, 但我们将其留作未来工作。

with initial promising results, but learning from more high-quality human annotations will be the desiderata to further improve the performance. Scaling up ReAct with multi-task training and combining it with complementary paradigms like reinforcement learning could result in stronger agents that further unlock the potential of LLMs for more applications.

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REPRODUCIBILITY STATEMENT

Our main experiments are done on PaLM (Chowdhery et al., 2022), which is not an openly accessible model yet. To increase reproducibility, we have included all used prompts in Appendix C, additional experiments using GPT-3 (Brown et al., 2020) in Appendix A.1, and associated GPT-3 ReAct prompting code at <https://anonymous.4open.science/r/ReAct-2268/>.

ETHICS STATEMENT

ReAct prompts large language models to generate more human interpretable, diagnosable, and controllable task-solving trajectories than previous methods. However, hooking up a large language model with an action space to interact with external environments (e.g. the web, physical environments) has potential dangers, e.g. looking up inappropriate or private information, or taking harmful actions in an environment. Our experiments minimize such risks by limiting the interactions to specific websites (Wikipedia or WebShop) that are free of private information, without any dangerous actions in the action space design (i.e. models cannot really buy products on WebShop the research benchmark, or edit Wikipedia). We believe researchers should be aware of such risks before designing more extensive experiments in the future.

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，并取得了初步的积极成果，但从更多高质量的人工标注中学习将是进一步提高性能的期望。通过多任务训练扩展ReAct，并将其与强化学习等互补范式相结合，可能会产生更强的智能体，从而进一步释放LLM在更多应用中的潜力。

致谢

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可复现性声明

我们的主要实验是在PaLM（Chowdhery等人，2022年）上完成的，该模型目前尚未公开。为提高可复现性，我们在附录C中包含了所有使用的提示，在附录A.1中提供了使用GPT-3（Brown等人，2020年）进行的附加实验，并在<https://anonymous.4open.science/r/ReAct-2268/>提供了相关的GPT-3 ReAct提示代码。

伦理声明

ReAct提示大型语言模型生成比以往方法更符合人类可解释性、可诊断性和可控性的任务解决轨迹。然而，将大型语言模型与动作空间连接起来以与外部环境（例如网络、物理环境）交互存在潜在危险，例如查询不适当或私密信息，或在环境中采取有害行动。我们的实验通过将交互限制在特定网站（维基百科或WebShop）来最小化此类风险，这些网站不包含私密信息，且动作空间设计（即模型无法在WebShop研究基准上真正购买产品，或编辑维基百科）中没有任何危险行动。我们相信研究人员在进行更广泛的实验之前应意识到此类风险。

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A ADDITIONAL RESULTS

A.1 GPT-3 EXPERIMENTS

	PaLM-540B	GPT-3
HotpotQA (exact match)	29.4	30.8
ALFWorld (success rate %)	70.9	78.4

Table 5: ReAct prompting results using PaLM-540B vs. GPT-3 (text-davinci-002, greedy decoding). On HotpotQA, we randomly sample a subset of 500 validation questions. On ALFWorld, we use all 134 unseen validation task instances, and use the best prompt set according to PaLM-540B.

We run additional GPT-3 (Brown et al., 2020) experiments to confirm ReAct prompting performance is general across different large language models. As shown in Table 5, GPT-3 (text-davinci-002, greedy decoding) consistently outperforms PaLM-540B on HotpotQA and ALFWorld, possibly because it is finetuned with human instruction following. This indicates ReAct prompting is effective across different large language models on different tasks. The code for these experiments are at <https://react-lm.github.io/>.

A.2 REACT OBTAINS UP-TO-DATE KNOWLEDGE ON HOTPOTQA

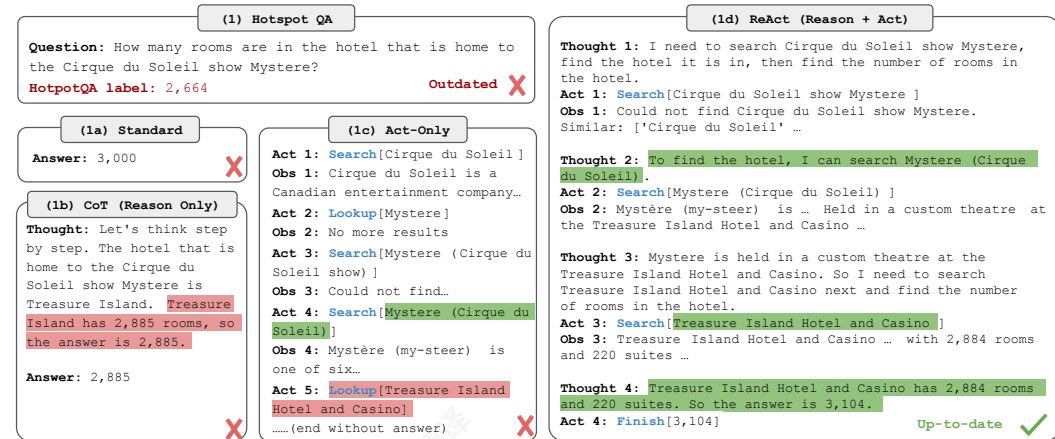


Figure 4: Another example HotpotQA question, where the original label is outdated. Only ReAct is able to obtain the up-to-date answer thanks to real-world web interaction plus reasoning.

During trajectory inspection, we also find that sometimes ReAct does not agree with dataset labels as the labels themselves could be outdated. For example, as shown in Figure 4, the question asks about the size of a hotel, which increased from the HotpotQA construction time. While Standard and CoT give wrong answers due to hallucination, Act fails despite the access of real-world web interaction, due to a lack of reasoning to guide how to interact with the Internet for QA. Only ReAct is able to retrieve up-to-date information from the Internet and provide a reasonable answer. Therefore, better incorporation of reasoning abilities might benefit recent Internet-augmented language models (Nakano et al., 2021; Lazaridou et al., 2022; Shuster et al., 2022a) for up-to-date task solving.

A.3 HUMAN-IN-THE-LOOP BEHAVIOR CORRECTION ON ALFWORLD

We also explore human-in-the-loop interaction with ReAct, to allow a human to inspect and edit ReAct’s reasoning traces. Figure 5 shows that by simply removing a hallucinating sentence in Act 17 and adding some hints in Act 23, ReAct can be made to change its behavior drastically to align with these human thought edits and succeed in the task. From a human perspective, solving such a task becomes significantly easier, from typing tens of actions to only editing a couple of thoughts, which enables new forms of human-machine collaboration. We note that such a policy edit on-the-go

附加结果

A.1 GPT-3实验

	PaLM-540B	GPT-3
HotpotQA (精确匹配)	29.4	30.8
ALFWorld (成功率%)	70.9	78.4

表5: 使用PaLM-540B与GPT-3 (text-davinci-002, 贪婪解码) 的ReAct提示结果。在HotpotQA上, 我们随机抽取500个验证问题子集。在ALFWorld上, 我们使用全部134个未见过的验证任务实例, 并使用根据PaLM-540B选择的最佳提示集。

我们运行额外的GPT-3 (Brown等人, 2020) 实验, 以确认ReAct提示性能在不同的大型语言模型上具有普遍性。如表5所示, GPT-3 (text-davinci-002, 贪婪解码) 在HotpotQA和ALFWorld上始终优于PaLM-540B, 可能因为它经过人类指令跟随的微调。这表明ReAct提示在不同的大型语言模型和不同任务上都是有效的。这些实验的代码位于 <https://react-lm.github.io/>。

A.2 ReAct在HotpotQA上获取最新知识

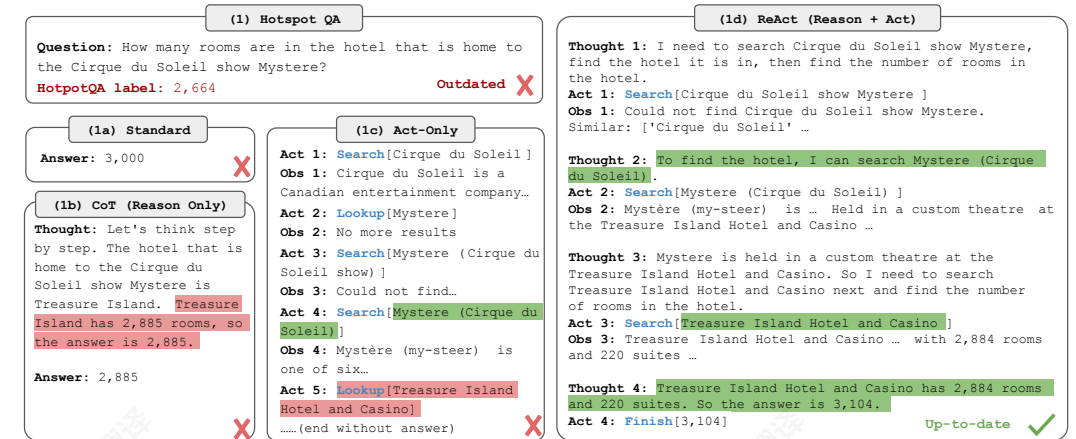


图4: 另一个HotpotQA问题示例, 其中原始标签已过时。只有ReAct能够通过现实世界的网络交互和推理获得最新的答案。

在轨迹检查过程中, 我们还发现ReAct有时与数据集标签不一致, 因为标签本身可能已过时。例如, 如图4所示, 问题询问ahotel的大小, 该大小自HotpotQA构建时间以来已有所增加。标准方法 (Standard) 和CoT由于幻觉 (hallucination) 给出错误答案, Act尽管能够访问现实世界的网络交互, 但由于缺乏推理来指导如何通过互联网进行问答, 仍然失败。只有ReAct能够从互联网检索最新信息并提供合理答案。因此, 更好地结合推理能力可能有助于近期互联网增强型语言模型 (Nakano等人, 2021; Lazaridou等人, 2022; Shuster等人, 2022a) 解决最新任务。

A.3 人类参与循环行为校正 ONALFWORLD

我们还探索了人类在环中的交互方式, 允许人类检查和编辑 ReAct 的推理轨迹。图 5 显示, 只需在 Act 17 中移除一句幻觉句子, 并在 Act 23 中添加一些提示, ReAct 就能大幅改变其行为以符合人类思维编辑, 从而成功完成任务。从人类的角度来看, 解决此类任务变得显著更容易, 从需要输入数十个动作到只需编辑几个想法, 这为新型人机协作提供了可能。我们注意到, 这种实时策略编辑对 Act 和之前的强化学习方法来说非常困难, 因为人类无法改变模型参数, 而修改少数几个动作可能无法编辑模型的其他行为。这种范式也不仅仅是像 Huang 等人 (2022b) 中那样通过人类对话来更新目标或子目标——虽然编辑 ReAct 思维可以做到这些, 但它还可以修改模型的内部信念、推理风格或灵活思维空间支持的其他任何内容, 以更好地解决任务。我们相信这是人类对齐领域的一个令人兴奋的方向, 并将更多系统性的研究留作未来工作。

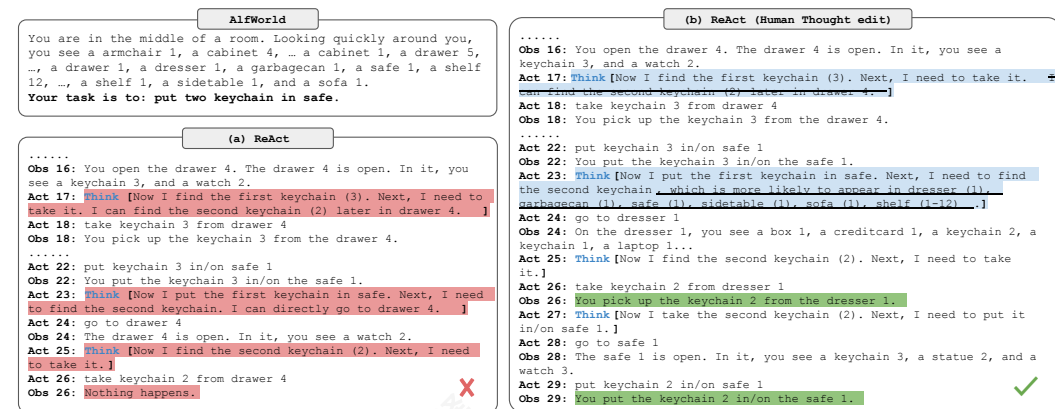


Figure 5: A human-in-the-loop behavior correction example with ReAct in AlfWorld. (a) ReAct trajectory fails due to a hallucinating thought (Act 17). (b) By a human simply editing two thoughts (Act 17, 23), the ReAct trajectory produces desirable reasoning traces and actions and succeeds.

is difficult for Act and previous RL methods, as a human cannot change the model parameters, and changing a few actions might not edit the rest of the model behavior. This paradigm is also more than human dialogue to update the goal or subgoal as in Huang et al. (2022b) — while editing ReAct thoughts can do these, it can also modify the model’s internal belief, reasoning styles, or anything the flexible thought space supports, for better task solving. We believe this is an exciting direction for human alignment and leave more systematic study as future work.

B EXPERIMENT DETAILS

B.1 HOTPOTQA FINETUNING DETAILS

For all finetuning we use a batch size of 64. On PaLM-8B, we finetune ReAct and Act methods for 4,000 steps and Standard and CoT methods for 2,000 steps. On PaLM-62B, we finetune ReAct and Act methods for 4,000 steps and Standard and CoT methods for 1,000 steps. We find ReAct and Act methods generally benefit from more training steps (and more training data), while Standard and CoT methods degrade soon after finetuning.

B.2 ALFWORLD IM-STYLE DETAILS

For the IM-style ablation, the same expert trajectories used in ReAct are reannotated with dense external feedback thoughts within these trajectories, that limit ReAct-IM to only think about (1) decomposing the current goal and (2) the current subgoal that needs to be completed. In particular, ReAct-IM lacks thoughts that (1) determine when a subgoal is completed (2) determine what the next subgoal should be (3) inducing the LLM to refer to its internal pretraining knowledge to identify where items can be within the environment.

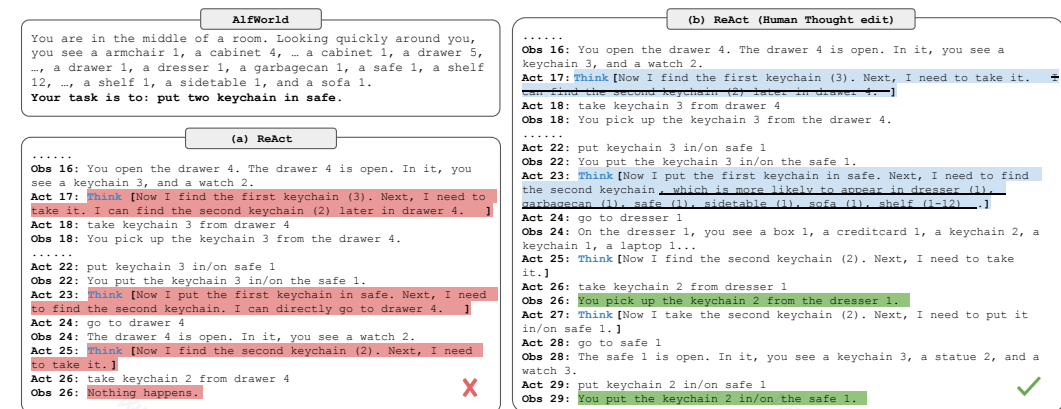


图5: ReAct在AlfWorld中的人回路行为校正示例。 (a) ReAct轨迹因产生幻觉性想法（行动17）而失败。 (b) 通过人类简单编辑两个想法（行动17、23），ReAct轨迹产生理想的推理轨迹和行动并成功。

这种策略编辑对 Act 和之前的强化学习方法来说非常困难，因为人类无法改变模型参数，而修改少数几个动作可能无法编辑模型的其他行为。这种范式也不仅仅是像 Huang 等人 (2022b) 中那样通过人类对话来更新目标或子目标——虽然编辑 ReAct 思维可以做到这些，但它还可以修改模型的内部信念、推理风格或灵活思维空间支持的其他任何内容，以更好地解决任务。我们相信这是人类对齐领域的一个令人兴奋的方向，并将更多系统性的研究留作未来工作。

B 实验细节

B.1 HOTPOTQA 微调细节

所有微调我们都使用批量大小为64。在PaLM-8B上，我们微调ReAct和Act方法4,000步，微调Standard和CoT方法2,000步。在PaLM-62B上，我们微调ReAct和Act方法4,000步，微调Standard和CoT方法1,000步。我们发现ReAct和Act方法通常受益于更多训练步（和更多训练数据），而Standard和CoT方法在微调后很快就会退化。

B.2 ALFWORLD IM-风格详情

对于IM风格的消融实验，ReAct中使用的相同专家轨迹会在这些轨迹内部重新标注密集的外部反馈思维，这些思维限制了ReAct-IM只能思考(1)分解当前目标以及(2)需要完成的当前子目标。具体来说，ReAct-IM缺乏(1)判断子目标何时完成(2)确定下一个子目标应该是什么(3)引导LLM参考其内部预训练知识以识别物品在环境中的位置的思维。

C PROMPTS

C.1 HOTPOTQA

Hotpot QA Prompts	
Original	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Answer	1,800 to 7,000 ft
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Answer	Richard Nixon
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Answer	The Saimaa Gesture
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Answer	director, screenwriter, actor
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Answer	Arthur’s Magazine
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Answer	Yes
Act	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Action 1	Search[Colorado orogeny]
Observation 1	The Colorado orogeny was an episode of mountain building (an orogeny) in Colorado and surrounding areas.
Action 2	Lookup[eastern sector]
Observation 2	(Result 1 / 1) The eastern sector extends into the High Plains and is called the Central Plains orogeny.
Action 3	Search[High Plains]
Observation 3	High Plains refers to one of two distinct land regions:
Action 4	Search[High Plains (United States)]
Observation 4	The High Plains are a subregion of the Great Plains. From east to west, the High Plains rise in elevation from around 1,800 to 7,000 ft (550 to 2,130 m).[3]
Action 5	Finish[1,800 to 7,000 ft]
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Action 1	Search[Milhouse]
Observation 1	Milhouse Mussolini Van Houten is a recurring character in the Fox animated television series The Simpsons voiced by Pamela Hayden and created by Matt Groening.
Action 2	Lookup[named after]
Observation 2	(Result 1 / 1) Milhouse was named after U.S. president Richard Nixon, whose middle name was Milhous.
Action 3	Finish[Richard Nixon]
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Action 1	Search[Adam Clayton Powell]
Observation 1	Could not find [Adam Clayton Powell]. Similar: [‘Adam Clayton Powell III’, ‘Seventh Avenue (Manhattan)’, ‘Adam Clayton Powell Jr. State Office Building’, ‘Isabel Washington Powell’, ‘Adam Powell’, ‘Adam Clayton Powell (film)’, ‘Giancarlo Esposito’].
Action 2	Search[Adam Clayton Powell (film)]
Observation 2	Adam Clayton Powell is a 1989 American documentary film directed by Richard Kilberg. The film is about the rise and fall of influential African-American politician Adam Clayton Powell Jr.[3][4] It was later aired as part of the PBS series The American Experience.
Action 3	Finish[The Saimaa Gesture]
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Action 1	Search[Nicholas Ray]
Observation 1	Nicholas Ray (born Raymond Nicholas Kienzle Jr., August 7, 1911 – June 16, 1979) was an American film director, screenwriter, and actor best known for the 1955 film Rebel Without a Cause.
Action 2	Search[Elia Kazan]
Observation 2	Elia Kazan was an American film and theatre director, producer, screenwriter and actor.

Continued on next page

C 提示

C.1 HOTPOTQA

Hotpot QA 提示	
原始	
问题	该地区的东部区域，科罗拉多造山运动延伸至的范围是多少？
答案	科罗拉多造山运动延伸至的范围是多少？ 1,800至7,000英尺
问题	音乐家和讽刺作家Allie Goertz创作了一首关于《辛普森一家》的歌曲角色米洛斯，马特·格勒宁是根据谁的名字命名的？
答案	理查德·尼克松
问题	哪部纪录片是关于芬兰摇滚乐队的，亚当·克莱顿·鲍威尔还是《Saimaa Gesture？
答案	萨伊马手势
问题	尼古拉斯·雷和伊利亚·卡赞有什么共同的职业？
答案	导演，编剧、演员
问题	《亚瑟杂志》和《首先为女性》哪本杂志创办时间更早？
答案	《亚瑟杂志》
问题	帕维尔·乌里松和列昂尼德·列文是否以相同类型的工作而闻名？
答案	Yes
Act	
问题	东部区域中科罗拉多造山运动延伸至的区域的海拔范围是多少？
操作 1	Search[Colorado orogeny]
观测 1	科罗拉多造山运动是科罗拉多州及周边地区的一次山脉建造事件（造山运动）。
操作 2	查找 [东部区域]
观测 2	(结果 1 / 1) 东部区域延伸至高平原，被称为中央平原造山带。
操作 3	搜索 [高平原]
观察3	High Plains 指的是两个不同的陆地区域之一：
操作4	搜索 [High Plains (美国)]
观察4	High Plains 是大平原的一个子区域。从东到西， 高平原海拔从约1,800英尺到7,000英尺（550米到2,130米）上升。[3]
行动5	完成 [1,800 到7,000英尺]
问题	音乐家和讽刺作家Allie Goertz创作了一首关于《辛普森一家》的歌 character Milhouse, who Matt Groening named after who?
操作 1	搜索 [米尔豪斯]
观察 1	米尔豪斯·墨索里尼·范·霍顿是福克斯动画电视系列《辛普森一家》中的一个常驻角色，由帕米拉·海登配音，由马特·格勒宁创作。
操作 2	查找 [以]命名
观察2	(结果1 / 1) Milhouse这个名字是为了纪念美国总统理查德·尼克松，他的中间名是Milhous。
操作3	完成 [理查德·尼克松]
问题	哪部纪录片是关于芬兰摇滚乐队的，亚当·克莱顿·鲍威尔还是The Saimaa手势？
操作 1	搜索 [Adam Clayton Powell]
观察 1	找不到 [亚当·克莱顿·鲍威尔]。Similar: [‘Adam Clayton Powell III’, ‘第七大道 (曼哈顿)’, ‘Adam Clayton Powell Jr. State Office Building’, ‘Isabel Washington Powell’, ‘Adam Powell’, ‘Adam Clayton Powell (film)’, ‘Giancarlo Esposito’].
动作2	搜索 [Adam Clayton Powell (电影)]
观察2	Adam Clayton Powell是一部1989年美国纪录片，由理查德·基尔伯格执导。这部影片讲述了有影响力的非裔美国政治家亚当·克莱顿·鲍威尔的崛起与沉浮。
行动3	完成 [萨伊马手势]
问题	尼古拉斯·雷和伊利亚·卡赞有什么共同的职业？
操作 1	搜索 [尼古拉斯·雷]
观察 1	尼古拉斯·雷（生于雷蒙德·尼古拉斯·凯恩勒，1911年8月7日 - 1979年6月16日）是一位美国电影导演、编剧和演员，以……闻名。
操作 2 观察 2	the 1955 film Rebel Without a Cause Search[Elia Kazan] Elia Kazan was an American film and theatre director, producer, screenwriter and actor.

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Hotpot QA Prompts – Continued from previous page	
Action 3	Finish[director, screenwriter, actor]
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Action 1	Search[Arthur’s Magazine]
Observation 1	Arthur’s Magazine (1844-1846) was an American literary periodical published in Philadelphia in the 19th century.
Action 2	Search[First for Women]
Observation 2	First for Women is a woman’s magazine published by Bauer Media Group in the USA.[1] The magazine was started in 1989.
Action 3	Finish[Arthur’s Magazine]
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Action 1	Search[Pavel Urysohn]
Observation 1	Pavel Samuilovich Urysohn (February 3, 1898 - August 17, 1924) was a Soviet mathematician who is best known for his contributions in dimension theory.
Action 2	Search[Leonid Levin]
Observation 2	Leonid Anatolievich Levin is a Soviet-American mathematician and computer scientist.
Action 3	Finish[yes]
CoT	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Thought	Let’s think step by step. The eastern sector of Colorado orogeny extends into the High Plains. High Plains rise in elevation from around 1,800 to 7,000 ft, so the answer is 1,800 to 7,000 ft.
Answer	1,800 to 7,000 ft
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Thought	Let’s think step by step. Milhouse was named after U.S. president Richard Nixon, so the answer is Richard Nixon.
Answer	Richard Nixon
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Thought	Let’s think step by step. Adam Clayton Powell (film) is a documentary about an African-American politician, not Finnish rock groups. So the documentary about Finnish rock groups must instead be The Saimaa Gesture.
Answer	The Saimaa Gesture
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Thought	Let’s think step by step. Professions of Nicholas Ray are director, screenwriter, and actor. Professions of Elia Kazan are director, producer, screenwriter, and actor. So profession Nicholas Ray and Elia Kazan have in common is director, screenwriter, and actor.
Answer	director, screenwriter, actor
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Thought	Let’s think step by step. Arthur’s Magazine was started in 1844. First for Women was started in 1989. 1844 (Arthur’s Magazine) < 1989 (First for Women), so Arthur’s Magazine was started first.
Answer	Arthur’s Magazine
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Thought	Let’s think step by step. Pavel Urysohn is a mathematician. Leonid Levin is a mathematician and computer scientist. So Pavel Urysohn and Leonid Levin have the same type of work.
Answer	Yes
ReAct	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Thought 1	I need to search Colorado orogeny, find the area that the eastern sector of the Colorado orogeny extends into, then find the elevation range of the area.
Action 1	Search[Colorado orogeny]
Observation 1	The Colorado orogeny was an episode of mountain building (an orogeny) in Colorado and surrounding areas.
Thought 2	It does not mention the eastern sector. So I need to look up eastern sector.
Action 2	Lookup[eastern sector]
Observation 2	(Result 1 / 1) The eastern sector extends into the High Plains and is called the Central Plains orogeny.
Thought 3	The eastern sector of Colorado orogeny extends into the High Plains. So I need to search High Plains and find its elevation range.
Action 3	Search[High Plains]
Observation 3	High Plains refers to one of two distinct land regions
Thought 4	I need to instead search High Plains (United States).
Action 4	Search[High Plains (United States)]

Continued on next page

火锅问答提示——接续上一页	
行动3	完成 [导演、编剧、演员]
问题	《亚瑟杂志》和《第一女人》哪本杂志创办时间更早？
操作 1	搜索 [《亚瑟杂志》]
观察 1	《亚瑟杂志》（1844-1846）是美国19世纪在费城出版的一种文学期刊。
操作 2	搜索 [《妇女第一》]
观察2	《First for Women》是一本由美国宝儿传媒集团出版的女性杂志。[1] 该杂志成立于1989年。
行动3	完成 [《亚瑟杂志》]
问题	帕维尔·乌里松和列昂尼德·列文以相同类型的工作而闻名吗？
操作 1	搜索 [帕维尔·乌里松]
观察 1	帕维尔·萨穆伊洛维奇·乌里松（1898年2月3日 - 1924年8月17日）是一位苏联数学家，以其在维数理论方面的贡献而闻名。
操作 2	搜索 [列昂尼德·列文]
观测2	列昂尼德·阿纳托利耶维奇·列文是一位苏联裔美国数学家和计算机科学家。
操作3	完成 [是]
CoT	
问题	科罗拉多山脉东部区域的海拔范围是多少？
思考	geny 延伸至？ 让我们一步步思考。科罗拉多山脉的东部延伸至大平原。大平原的海拔从约1,800英尺到7,000英尺，所以答案是1,800英尺到7,000英尺。
答案	1,800英尺到7,000英尺
问题	音乐家和讽刺作家艾莉·格茨写了一首歌关于《辛普森一家》角色米尔豪斯，马特·格莱宁给这个角色命名是为了谁？
思考	让我们一步步思考。米尔豪斯是以美国总统理查德·尼克松命名的，所以答案是理查德·尼克松。
答案	理查德·尼克松
问题	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa手势？
Thought	让我们一步步思考。亚当·克莱顿·鲍威尔（电影）是一部关于非裔美国政治家的纪录片，而不是芬兰摇滚乐队。所以关于芬兰摇滚乐队的纪录片必须是《萨伊马手势》。
回答	The Saimaa Gesture
问题	尼古拉斯·雷和伊利亚·卡赞有什么职业共同点？
思考	让我们一步步思考。尼古拉斯·雷的职业是导演、编剧和演员。伊利亚·卡赞的职业是导演、制片人、编剧和演员。所以尼古拉斯·雷和伊利亚·卡赞有共同的职业是导演、编剧和演员。
答案	共同点是导演、编剧和演员 导演、编剧、演员
问题	《亚瑟杂志》和《首先为女性》哪本杂志创办得早？
思考	让我们一步步思考。亚瑟杂志创办于1844年，首先为女性创办于1989年。1844年（亚瑟杂志）< 1989 1989年（首先为女性），所以亚瑟杂志创办得早。
答案	亚瑟杂志
问题	帕维尔·乌里松和列昂尼德·列文是否以相同类型的工作而闻名？
思考	让我们一步步思考。帕维尔·乌里松是一位数学家。列昂尼德·列文是一位数学家和计算机科学家。因此，帕维尔·乌里松和列昂尼德·列文从事相同类型的工作。
答案	Yes
ReAct	
问题	该区域的东部区域的海拔范围是多少？
思考 1	科罗拉多造山运动延伸至？ 我需要搜索科罗拉多造山运动，找到东部区域的范围 of the Colorado orogeny extends into, then find the elevation range of the area.
行动 1	Search[Colorado orogeny]
观测1	The Colorado orogeny was an episode of mountain building (an orogeny) in 科罗拉多州及周边地区 g 地区 .
思考2	它没有提到东部地区。 所以我需要查找东部区域。
操作 2	查找 [东部区域]
观测 2	(结果 1 / 1) 东部区域延伸至高平原，被称为中央平原造山带 .
思考3	科罗拉多造山运动的东西部延伸至 高平原。 So I 需要搜索高平原并找到其海拔范围。
行动3	搜索 [高平原]
观察3	高平原指两个不同的陆地区域之一
思考4	我需要改为搜索高平原（美国）。
行动4	搜索 [高平原（美国）]

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Hotpot QA Prompts – Continued from previous page	
Observation 4	The High Plains are a subregion of the Great Plains. From east to west, the High Plains rise in elevation from around 1,800 to 7,000 ft (550 to 2,130 m).[3]
Thought 5	High Plains rise in elevation from around 1,800 to 7,000 ft, so the answer is 1,800 to 7,000 ft.
Action 5	Finish[1,800 to 7,000 ft]
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Thought 1	The question simplifies to "The Simpsons" character Milhouse is named after who. I only need to search Milhouse and find who it is named after.
Action 1	Search[Milhouse]
Observation 1	Milhouse Mussolini Van Houten is a recurring character in the Fox animated television series The Simpsons voiced by Pamela Hayden and created by Matt Groening.
Thought 2	The paragraph does not tell who Milhouse is named after, maybe I can look up "named after".
Action 2	Lookup[named after]
Observation 2	(Result 1 / 1) Milhouse was named after U.S. president Richard Nixon, whose middle name was Milhous.
Thought 3	Milhouse was named after U.S. president Richard Nixon, so the answer is Richard Nixon.
Action 3	Finish[Richard Nixon]
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Thought 1	I need to search Adam Clayton Powell and The Saimaa Gesture, and find which documentary is about Finnish rock groups.
Action 1	Search[Adam Clayton Powell]
Observation 1	Could not find [Adam Clayton Powell]. Similar: ['Adam Clayton Powell III', 'Seventh Avenue (Manhattan)', 'Adam Clayton Powell Jr. State Office Building', 'Isabel Washington Powell', 'Adam Powell', 'Adam Clayton Powell (film)', 'Giancarlo Esposito'].
Thought 2	To find the documentary, I can search Adam Clayton Powell (film).
Action 2	Search[Adam Clayton Powell (film)]
Observation 2	Adam Clayton Powell is a 1989 American documentary film directed by Richard Kilberg. The film is about the rise and fall of influential African-American politician Adam Clayton Powell Jr.[3][4] It was later aired as part of the PBS series The American Experience.
Thought 3	Adam Clayton Powell (film) is a documentary about an African-American politician, not Finnish rock groups. So the documentary about Finnish rock groups must instead be The Saimaa Gesture.
Action 3	Finish[The Saimaa Gesture]
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Thought 1	I need to search Nicholas Ray and Elia Kazan, find their professions, then find the profession they have in common.
Action 1	Search[Nicholas Ray]
Observation 1	Nicholas Ray (born Raymond Nicholas Kienzle Jr., August 7, 1911 – June 16, 1979) was an American film director, screenwriter, and actor best known for the 1955 film Rebel Without a Cause.
Thought 2	Professions of Nicholas Ray are director, screenwriter, and actor. I need to search Elia Kazan next and find his professions.
Action 2	Search[Elia Kazan]
Observation 2	Elia Kazan was an American film and theatre director, producer, screenwriter and actor.
Thought 3	Professions of Elia Kazan are director, producer, screenwriter, and actor. So profession Nicholas Ray and Elia Kazan have in common is director, screenwriter, and actor.
Action 3	Finish[director, screenwriter, actor]
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Thought 1	I need to search Arthur’s Magazine and First for Women, and find which was started first.
Action 1	Search[Arthur’s Magazine]
Observation 1	Arthur’s Magazine (1844-1846) was an American literary periodical published in Philadelphia in the 19th century.
Thought 2	Arthur’s Magazine was started in 1844. I need to search First for Women next.
Action 2	Search[First for Women]
Observation 2	First for Women is a woman’s magazine published by Bauer Media Group in the USA.[1] The magazine was started in 1989.
Thought 3	First for Women was started in 1989. 1844 (Arthur’s Magazine) < 1989 (First for Women), so Arthur’s Magazine was started first.
Action 3	Finish[Arthur’s Magazine]
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Thought 1	I need to search Pavel Urysohn and Leonid Levin, find their types of work, then find if they are the same.
Action 1	Search[Pavel Urysohn]

火锅问答提示——接续上一页	
观察4	高平原是大平原的一个子区域。从东到西，高平原的海拔从约1,800英尺到7,000英尺（550米到2,130米）上升。[3]
思考5	高平原海拔从约1,800英尺上升到7,000英尺，所以答案是1,800到7,000英尺。
行动5	完成[1,800 到7,000英尺]
问题	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
思考1	这个问题简化为《辛普森一家》角色米洛斯的名字是根据谁来的。
行动1	who 我只需要搜索米洛斯，找出它名字是根据谁来的。
观察1	搜索[米洛斯] Milhouse Mussolini Van Houten is a recurring character in the Fox animated television series The Simpsons voiced by Pamela Hayden and created by Matt Groening.
思考 2	The paragraph does not tell who Milhouse is named after, maybe I can look up "named after".
行动 2	查找[以命名
观察 2	(结果 1 / 1) Milhouse以美国总统理查德·尼克松命名，其中间名为Milhous。
思考 3	米尔豪斯的名字是为了纪念美国总统理查德·尼克松，所以答案是理查德·尼克松。
行动3	完成[理查德·尼克松]
问题	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
思考 1	我需要搜索亚当·克莱顿·鲍威尔，纪录片是关于芬兰摇滚乐队的。The Saimaa Gesture, and find which
行动 1	搜索[亚当·克莱顿·鲍威尔]
观察 1	未找到 [亚当·克莱顿·鲍威尔]。相似: ['亚当·克莱顿·鲍威尔 III', 'Seventh Avenue (Manhattan)', 亚当·克莱顿·鲍威尔小省办公厅建筑', '伊莎贝尔·华盛顿·鲍威尔', '亚当·鲍威尔', '亚当·克莱顿·鲍威尔 (电影)', '吉安卡洛·埃斯波西托']。
思考2	要找到纪录片，我可以搜索亚当·克莱顿·鲍威尔（电影）。
行动2	搜索[亚当·克莱顿·鲍威尔（电影）]
观察2	Adam Clayton Powell is a 1989 American documentary film directed by 理查德·基尔伯 g. 这部电影讲述了重要人物的兴衰
思想3	African-American politician Adam Clayton Powell Jr.[3][4] It was later aired as part of the PBS series The American Experience.
行动3	Adam Clayton Powell (film) is a documentary about an African-American politician, not Finnish rock groups. So the documentary about Finnish rock groups must instead be The Saimaa Gesture.
完成	完成[萨伊马手势]
问题	尼古拉斯·雷和伊利亚·卡赞有什么职业共同点?
想法1	我需要搜索尼古拉斯·雷和伊利亚·卡赞，找到他们的职业，然后找到他们共同的职业。
操作1	搜索[尼古拉斯·雷]
观察1	尼古拉斯·雷（本名雷蒙德·尼古拉斯·凯恩斯勒，1911年8月7日—1979年6月16日）是一位美国电影导演、编剧和演员，以1955年电影《无因的反抗》闻名。
思考2	尼古拉斯·雷的职业是导演、编剧和演员。我需要接下来搜索伊利亚·卡赞，并找到他的职业。
行动2	搜索[伊利亚·卡赞]
观察2	伊利亚·卡赞是一位美国电影和戏剧导演、制片人、编剧和演员。
思考3	伊利亚·卡赞的职业是导演、制片人、编剧和演员。所以尼古拉斯·雷和伊利亚·卡赞的共同职业是导演、编剧和演员。
行动3	完成[导演、编剧、演员]
问题	《亚瑟杂志》和《第一夫人》哪本创办时间更早?
思考 1	我需要搜索《亚瑟杂志》和《第一夫人》，并找出哪本创办时间更早。
行动 1	搜索[亚瑟杂志]
观察1	亚瑟杂志（1844-1846）是美国19世纪在费城出版的一种文学期刊。
思考2	亚瑟杂志于1844年创办。我需要先搜索《妇女》
操作 2	下一位。
观察 2	为女性搜索[, 首先] 《首先为女性》是一本由鲍尔传媒集团出版的女性杂志, USA.[1]该杂志始于1989年。
思考3	First for Women was started in 1989. 1844 (《亚瑟杂志》) < 1989 (首次为女性), 因此《亚瑟杂志》是首先创办的。
行动3	完成[《亚瑟杂志》]
问题	Were Pavel Urysohn and Leonid Levin known for the same type of work?
思考 1	I need to search Pavel Urysohn and Leonid Levin, find their types of work, then find if they are the same.
行动 1	Search[Pavel Urysohn]

Hotpot QA Prompts – Continued from previous page	
Observation 1	Pavel Samuilovich Urysohn (February 3, 1898 – August 17, 1924) was a Soviet mathematician who is best known for his contributions in dimension theory.
Thought 2	Pavel Urysohn is a mathematician. I need to search Leonid Levin next and find its type of work.
Action 2	Search[Leonid Levin]
Observation 2	Leonid Anatolievich Levin is a Soviet-American mathematician and computer scientist.
Thought 3	Leonid Levin is a mathematician and computer scientist. So Pavel Urysohn and Leonid Levin have the same type of work.
Action 3	Finish[yes]

火锅问答提示——接续上一页	
观察1	帕维尔·萨穆伊洛维奇·乌里松（1898年2月3日 - 1924年8月17日）是一位苏联数学家，以其在维数理论方面的贡献而闻名。
思考2	帕维尔·乌里松是一位数学家。我需要接下来搜索列昂尼德·列文，并找到其工作类型。
操作 2	搜索 [列昂尼德·列文]
观察 2	列昂尼德·阿纳托利耶维奇·列文是一位苏联裔美国数学家和计算机科学家。
思考 3	列昂尼德·列文是乌里松的同事和同僚，他们的工作领域相同。
行动3	完成 [是]

C.2 FEVER

FEVER Prompts	
Original	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim Answer	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company. SUPPORTS
Claim Answer	Stranger Things is set in Bloomington, Indiana. REFUTES
Claim Answer	Beautiful reached number two on the Billboard Hot 100 in 2003. NOT ENOUGH INFO
Ask	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim Action2 Observation 1	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company. Search[Nikolaj Coster-Waldau] Nikolaj William Coster-Waldau (born 27 July 1970) is a Danish actor and producer. He graduated from the Danish National School of Performing Arts in Copenhagen in 1993,[1] and had his breakthrough role in Denmark with the film Nightwatch (1994). He played Jaime Lannister in the HBO fantasy drama series Game of Thrones, for which he received two Primetime Emmy Award nominations for Outstanding Supporting Actor in a Drama Series.. Coster-Waldau has appeared in numerous films in his native Denmark and Scandinavia, including Headhunters (2011) and A Thousand Times Good Night (2013). In the U.S, his debut film role was in the war film Black Hawk Down (2001), playing Medal of Honor recipient Gary Gordon.[2] He then played a detective in the short-lived Fox television series New Amsterdam (2008), and appeared in the 2009 Fox television film Virtuality, originally intended as a pilot.
Action 2	Finish[SUPPORTS]
Claim Action 1 Observation 1	Stranger Things is set in Bloomington, Indiana. Search[Stranger Things] Stranger Things is an American science fiction horror drama television series created by the Duffer Brothers. Set in the 1980s, primarily in the fictional town of Hawkins, Indiana, the series centers on a number of mysteries and supernatural events occurring around the town and their impact on an ensemble of child and adult characters.
Action 2	Finish[REFUTES]
Claim Action 1 Observation 1	Beautiful reached number two on the Billboard Hot 100 in 2003. Search[Beautiful] Could not find [Beautiful]. Similar: ['Beautiful', 'Beautiful, Beautiful', 'A Beautiful Mind (film)', 'Beautiful (Christina Aguilera song)', 'Life Is Beautiful'].
Action 2 Observation 2	Search[Beautiful (Christina Aguilera song)] "Beautiful" is a song recorded by American singer Christina Aguilera for her fourth studio album, Stripped (2002).
Action 3 Observation 3	Lookup[Billboard Hot 100] (Result 1 / 3) The song peaked at number two on the Billboard Hot 100 in the United States, where it was certified Gold for 500,000 units shipped.
Action 4	Finish[NOT ENOUGH INFO]
Ask	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim Thought Answer	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company. Nikolaj William Coster-Waldau appeared in the 2009 Fox television film Virtuality, so he has worked with the Fox Broadcasting Company. SUPPORTS
Claim Thought Answer	Stranger Things is set in Bloomington, Indiana. Stranger Things is in the fictional town of Hawkins, Indiana, not in Bloomington, Indiana. REFUTES

Continued on next page

C.2 FEVER

FEVER提示	
原始	判断是否存在支持或反驳声明的观察，或是否存在信息不足。
声明回答	Nikolaj Coster-Waldau 与福克斯广播公司合作。支持
声明回答	《怪奇物语》设定在印第安纳州的布卢明顿。反驳
主张答案	Beautiful 在 2003 年登上了公告牌百强单曲榜第二名。信息不足
Ask	确定是否存在支持或反驳主张的观察信息不足。 , 或者如果
声明动作2观察1	尼科莱·科斯特-瓦尔道与福克斯广播公司合作。搜索 [尼科莱·科斯特-瓦尔道] Николай·威廉·科斯特-沃尔道（1970年7月27日出生）是一位丹麦演员和制片人。他毕业于哥本哈根的丹麦国家表演艺术学院，在丹麦凭借电影《夜班守卫》（1994年）获得突破性角色。他在HBO奇幻剧《权力的游戏》中饰演杰迈雅·兰尼斯特，并因此获得两项黄金时段艾美奖最佳剧情类配角提名。科斯特-沃尔道在他家乡丹麦和斯堪的纳维亚半岛出演了多部电影，包括《猎头者》（2011年）和《晚安千次》（2013年）。在美国，他的首部电影角色是在战争电影《黑鹰坠落》（2001年）中饰演获得荣誉勋章的加里·戈登。他后来在短命的福克斯电视台剧集《新阿姆斯特丹》（2008年）中饰演一名侦探，并在2009年福克斯电视台电影《虚拟性》（最初计划作为试播集）中出演。
操作2	完成 [支持]
由领操作1观察1	《怪奇物语》设定在印第安纳州的布卢明顿。搜索 [怪奇物语] 《怪奇物语》是一部由达弗兄弟创作的美国科幻恐怖剧情类电视剧。故事设定在20世纪80年代，主要发生在印第安纳州霍金斯这个虚构小镇，剧集围绕小镇发生的一系列谜团和超自然事件，以及这些事件对一群儿童和成人角色的影响展开。
行动2	完成 [反驳]
主张操作 1观察 1	Beautiful在2003年登上了公告牌百强单曲榜的亚军位置。搜索 [美丽] 找不到 [美丽]。类似: [' 美丽' , ' 美丽, 美丽' , ' 美丽心灵 (电影)' , ' 美丽的 (克里斯蒂娜·阿奎莱拉歌曲)' , ' 生活是' 美丽的']
操作 2 Observation 2	搜索 [美丽的 (克里斯蒂娜·阿奎莱拉的歌曲)] "美丽"是一首由美国歌手克里斯蒂娜·阿奎莱拉为她的第四张录音室专辑录制的歌曲，Stripped (2002).
动作3观察3	查找 [公告牌热门 100] (结果1 / 3) 这首歌在美国公告牌百强单曲榜上排名第二，销量达到50万张，获得金唱片认证。
动作4	完成 [信息不足]
Ask	确定是否存在支持或反驳声明的观察信息不足。 , 或如果
声明思考	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company. 尼古拉斯·威廉·科斯特-沃尔道在2009年福克斯电视台的电影《虚拟性》中出演，因此他与福克斯广播公司合作过。
答案	支持
主张思考	《怪奇物语》设定在印第安纳州的布卢明顿。《怪奇物语》设定在印第安纳州的霍金斯市，而不是在布卢明顿市。
答案	反驳

续页

Fever Prompts – Continued from previous page	
Claim	Beautiful reached number two on the Billboard Hot 100 in 2003.
Thought	The song peaked at number two on the Billboard Hot 100 in the United States, but not sure if it was in 2003.
Answer	NOT ENOUGH INFO
ReAct	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company.
Thought 1	I need to search Nikolaj Coster-Waldau and find if he has worked with the Fox Broadcasting Company.
Action 1	Search[Nikolaj Coster-Waldau]
Observation 1	Nikolaj William Coster-Waldau (born 27 July 1970) is a Danish actor and producer. He graduated from the Danish National School of Performing Arts in Copenhagen in 1993,[1] and had his breakthrough role in Denmark with the film Nightwatch (1994). He played Jaime Lannister in the HBO fantasy drama series Game of Thrones, for which he received two Primetime Emmy Award nominations for Outstanding Supporting Actor in a Drama Series.. Coster-Waldau has appeared in numerous films in his native Denmark and Scandinavia, including Headhunters (2011) and A Thousand Times Good Night (2013). In the U.S, his debut film role was in the war film Black Hawk Down (2001), playing Medal of Honor recipient Gary Gordon.[2] He then played a detective in the short-lived Fox television series New Amsterdam (2008), and appeared in the 2009 Fox television film Virtuality, originally intended as a pilot.
Thought 2	Because he "appeared in the 2009 Fox television film Virtuality", he should have worked with the Fox Broadcasting Company.
Action 2	Finish[SUPPORTS]
Claim	Stranger Things is set in Bloomington, Indiana.
Thought 1	I should search for Stranger Things, and see if it is set in Bloomington, Indiana.
Action 1	Search[Stranger Things]
Observation 1	Stranger Things is an American science fiction horror drama television series created by the Duffer Brothers. Set in the 1980s, primarily in the fictional town of Hawkins, Indiana, the series centers on a number of mysteries and supernatural events occurring around the town and their impact on an ensemble of child and adult characters.
Thought 2	The observation says that it is set in a "fictional town of Hawkins, Indiana", so it is not set in Bloomington.
Action 2	Finish[REFUTES]
Claim	Beautiful reached number two on the Billboard Hot 100 in 2003.
Thought 1	I need to search the song Beautiful and find if it reached number two on the Billboard Hot 100 in 2003.
Action 1	Search[Beautiful]
Observation 1	Could not find [Beautiful]. Similar: ['Beautiful', 'Beautiful, Beautiful', 'A Beautiful Mind (film)', 'Beautiful (Christina Aguilera song)', 'Life Is Beautiful'].
Thought 2	From suggestions, I should search "Beautiful (Christina Aguilera song)" to find the song.
Action 2	Search[Beautiful (Christina Aguilera song)]
Observation 2	"Beautiful" is a song recorded by American singer Christina Aguilera for her fourth studio album, Stripped (2002).
Thought 3	It does not mention Billboard, so I need to look up "Billboard Hot 100" to find if it reached number two on it in 2003.
Action 3	Lookup[Billboard Hot 100]
Observation 3	(Result 1 / 3) The song peaked at number two on the Billboard Hot 100 in the United States, where it was certified Gold for 500,000 units shipped.
Thought 4	It only says the song peaked at number two on the Billboard Hot 100, but not if it was in 2003. I am not sure if this claim is true or not.
Action 4	Finish[NOT ENOUGH INFO]

发烧提示——续自上一页	
主张	Beautiful在2003年登上了Billboard Hot 100排行榜第二名。
思考	这首歌在美国Billboard Hot 100排行榜上达到了第二名，但不确定是否是在2003年。
回答	信息不足
ReAct	确定是否存在支持或反驳声明的观察结果， 或者是否存在信息不足的情况
声明	尼科莱·科斯特-沃尔道与福克斯广播公司合作过。
想法 1	我需要搜索尼科莱·科斯特-沃尔道，看看他是否与福克斯广播公司合作过。
行动 1	搜索 [尼科莱·科斯特-沃尔道]
观察1	Николай·威廉·科斯特-沃尔道（1970年7月27日出生）是一位丹麦演员和制片人。他于 1993, [1] 毕业于哥本哈根的丹麦国立表演艺术学院，并在丹麦的电影《夜班守卫》（1994年）中获得了突破性角色。他在HBO奇幻剧《权力的游戏》中饰演杰迈玛·兰尼斯特，并因此获得了两次剧情类最佳男配角提名。
思考2	科斯特-沃尔道在他家乡丹麦和斯堪的纳维亚半岛出演了多部电影，包括《猎头者》（2011年）和《晚安千次》（2013年）。在美国，他的首部电影角色是在战争片《黑鹰坠落》（2001年）中饰演获得荣誉勋章的加里·戈登。[2] 随后，他在短命的福克斯电视系列《新阿姆斯特丹》（2008年）中饰演一名侦探，并出现在2009年福克斯电视电影《虚拟性》，最初计划作为试播集。
思考2	因为他“出演了2009年福克斯电视台的电影《虚拟性》”，所以应该与福克斯广播公司合作过。
行动2	完成 [支持]
主张	《怪奇物语》设定在印第安纳州的布卢明顿。
想法 1	我应该搜索《怪奇物语》，看看它是否设定在印第安纳州的布卢明顿。
行动 1	搜索 [《怪奇物语》]
观察 1	《怪奇物语》是一部由达弗兄弟创作的美国科幻恐怖剧情类电视剧。故事设定在 20 世纪 80 年代，主要发生在印第安纳州虚构的霍金斯镇，该剧围绕镇上发生的一系列谜团和超自然事件，以及这些事件对一群儿童和成人角色的影响展开。
思考2	观察表明，它设定在一个名为“印第安纳州霍金斯镇”的虚构城镇中，因此不在布卢明顿市。
行动2	完成 [反驳]
主张	Beautiful在2003年登上了公告牌百强单曲榜第2名。
思考 1	我需要搜索歌曲《Beautiful》，并查看它在 2003 年是否登上了公告牌百强单曲榜第二名。
行动 1	搜索 [Beautiful]
观察 1	未找到 [Beautiful]。相似: [‘美丽’，‘美丽，美丽’，《美丽心灵》(电影) ‘美丽的 (克里斯蒂娜·阿奎莱拉歌曲)’，’ 生活是’美丽的’]
想法2	从建议中，我应该搜索“Beautiful (Christina Aguilera歌曲)”来找到这首歌。
操作 2	搜索 [美丽 (克里斯蒂娜·阿奎莱拉的歌曲)]
观察2	"美丽"是一首由美国歌手克里斯蒂娜·阿奎莱拉为她的第四张录音室专辑《Stripped》（2002年）录制的歌曲。
想法3	它没有提到公告牌，所以我需要查找“公告牌百强单曲榜”以确定它在2003年是否达到了第2名。
操作3	查询 [广告牌热榜 100]
观察3	(Result 1 / 3) 这首歌曲在美国的公告牌百强单曲榜上排名第二，并因其50万份的销量而获得黄金认证。
想法4	它只说这首歌在Billboard Hot 100上排名第二，但没有说明是2003年。我不确定这个说法是否属实。
操作4	完成 [信息不足]

C.3 WEBSHOP

Table 6: Prompts used for the Webshop experiments in Section 4. We use the prompt on the left for `Act`, and use the prompt on the right (add reasoning) for `ReAct`.

Instruction: i would like a 3 ounce bottle of bright citrus deodorant for sensitive skin, and price lower than 50.00 dollars	
Act	ReAct
Action: search [3 ounce bright citrus deodorant sensitive skin] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B078GWRC1J] Bright Citrus Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce \$10.99 [B078GTKVXY] Ginger Fresh Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce \$10.99 [B08KBVJ4XN] Barrel and Oak - Aluminum-Free Deodorant, Deodorant for Men , Essential Oil-Based Scent, 24-Hour Odor Protection, Cedar & Patchouli Blend, Gentle on Sensitive Skin (Mountain Sage, 2.7 oz, 2-Pack) \$15.95 Action: click [B078GWRC1J] Observation: [Back to Search] [Prev] scent [assorted scents][bright citrus][calming lavender][ginger [fresh] simply non-scents] size [travel set (4-pack)][3 ounce (pack of 1)][3-ounce (2 -pack)] Bright Citrus Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce Price: \$10.99 Rating: N.A. [Description] [Features] [Reviews] [Buy Now] Action: click[bright citrus] Observation: You have clicked bright citrus. Action: click[3 ounce (pack of 1)] Observation: You have clicked 3 ounce (pack of 1). Action: click[Buy Now]	Action: search [3 ounce bright citrus deodorant sensitive skin] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B078GWRC1J] Bright Citrus Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce \$10.99 [B078GTKVXY] Ginger Fresh Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce \$10.99 [B08KBVJ4XN] Barrel and Oak - Aluminum-Free Deodorant, Deodorant for Men , Essential Oil-Based Scent, 24-Hour Odor Protection, Cedar & Patchouli Blend, Gentle on Sensitive Skin (Mountain Sage, 2.7 oz, 2-Pack) \$15.95 Action: think[B078GWRC1J and B078GTKVXY are bright citrus deodorant less then 50 dollars. I can check B078GWRC1J first. Observation: OK. Action: click [B078GWRC1J] Observation: [Back to Search] [Prev] scent [assorted scents][bright citrus][calming lavender][ginger [fresh] simply non-scents] size [travel set (4-pack)][3 ounce (pack of 1)][3-ounce (2 -pack)] Bright Citrus Deodorant by Earth Mama Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce Price: \$10.99 Rating: N.A. [Description] [Features] [Reviews] [Buy Now] Action: think[For 3 ounce bottle of bright citrus deodorant for sensitive skin, the item has options 'bright citrus' and '3 ounce (pack of 1)' and seems good to buy.] Observation: OK. Action: click[bright citrus] Observation: You have clicked bright citrus. Action: click[3 ounce (pack of 1)] Observation: You have clicked 3 ounce (pack of 1). Action: click[Buy Now]

C.3 WEBSHOP

表6：第4节中Webshop实验使用的提示。我们对`Act`使用左侧提示，对`ReAct`（添加推理）使用右侧提示。

指令: 我想要一瓶3盎司的亮柑橘香氛止汗剂，适用于敏感肌肤，价格低于50.00美元	
Act	ReAct
Action: search [3 ounce bright citrus deodorant sensitive skin] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B078GWRC1J] 亮柑橘香氛止汗剂，由Earth Mama生产 天然且适用于敏感肌肤、孕妇及哺乳期，含有有机金盏花，3盎司装 \$ 10.99 [B078GTKVXY] 姜味清新止汗剂，由Earth Mama生产 天然且安全，适合敏感肌肤、孕妇及哺乳期使用，含有有机金盏花，3盎司装 \$10.99 [B08KBVJ4XN] Barrel and Oak - 无铝止汗剂，男士止汗剂，精油香型，24小时除臭保护，雪松与广藿香混合香型，温和不刺激敏感肌肤（山葵，2.7盎司，2件装） \$15.95 操作: 点击[B078GWRC1J] 观察: [返回搜索] [上一页] 香味 [各种香味][明亮的柑橘味][舒缓的薰衣草味][姜味 [清新味 纯粹的香味] 尺寸 [旅行套装（4件装）][3盎司（1件装）][3-盎司（2 -件装）] 地球妈妈亮柑橘除臭剂 天然安全，适合敏感肌肤、孕妇及哺乳期，含有有机金盏花，3盎司，价格: \$10.99 评分: 暂无 [Description] [Features] [Reviews] [Buy Now] Action: click[bright citrus] Observation: You have clicked bright citrus. Action: click[3 ounce (pack of 1)] Observation: You have clicked 3 ounce (pack of 1). 操作: 点击[立即购买]	Action: search [3 ounce bright citrus deodorant sensitive skin] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B078GWRC1J] 地球妈妈亮柑橘除臭剂 天然安全，适合敏感肌肤、孕妇及哺乳期，含有有机金盏花，3盎司 \$10.99 [B078GTKVXY] 姜味清新止汗剂，由 Earth Mama 生产 天然且安全，适合敏感肌肤、孕妇及哺乳期使用，含有有机金盏花，3盎司装 \$10.99 [B08KBVJ4XN] Barrel and Oak -无铝止汗剂，男士专用，精油香型，24小时除臭保护，雪松与广藿香混合香氛，温和不刺激敏感肌肤（山地鼠尾草，2.7盎司，2件装） \$15.95 操作: 思考[B078GWRC1J 和 B078GTKVXY 是两款价格低于50美元的柑橘味止汗剂。我可以先检查 B078GWRC1J。观察: OK。 Action: click [B078GWRC1J] Observation: [返回搜索] [上一页] scent [各种香气][明亮的柑橘香][舒缓的薰衣草香][姜味 [清爽 纯粹的香气] size [旅行套装（4件套）][3盎司（1包）][3-盎司（2 -{pack}}] <style>Bright Citrus除臭剂由EarthMama生产 天然且安全，适用于敏感肌肤、孕妇及哺乳期，含有有机金盏花 3盎司 价格: \$10.99 . . Rating: N A [Description] [Features] [Reviews] [Buy Now] 操作: 思考[对于3盎司装的亮丽柑橘味止汗剂，敏感肌肤适用，该商品有“亮丽柑橘”和“3盎司（1包）”的选项，看起来适合购买。] 观察: OK。 Action: click[bright citrus] Observation: You have clicked bright citrus. Action: click[3 ounce (pack of 1)] Observation: You have clicked 3 ounce (pack of 1). 操作: 点击[立即购买]

C.4 ALFWORLD

Table 7: An Act prompt on the ALFWorld clean task. No thoughts are provided.

You are in the middle of a room. Looking quickly around you, you see a cabinet 13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet 7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean lettuce in diningtable.

> go to fridge 1
The fridge 1 is closed.
> open fridge 1
You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a potato 3, and a potato 2.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1.
> take lettuce 1 from diningtable 1
You pick up the lettuce 1 from the diningtable 1.
> go to sinkbasin 1
On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3.
> clean lettuce 1 with sinkbasin 1
You clean the lettuce 1 using the sinkbasin 1.
> think: Now I clean a lettuce (1). Next, I need to put it in/on diningtable 1.
OK.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1.
> put lettuce 1 in/on diningtable 1
You put the lettuce 1 in/on the diningtable 1.

C.4 ALFWORLD

表7: ALFWorld干净任务上的一个行动提示。未提供任何思考。

You are in the middle of a room. Looking quickly around you, you see a cabinet 13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet 7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean lettuce in diningtable.

> go to fridge 1
The fridge 1 is closed.
> open fridge 1
You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a potato 3, and a potato 2.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1.
> take lettuce 1 from diningtable 1
You pick up the lettuce 1 from the diningtable 1.
> go to sinkbasin 1
On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3.
> clean lettuce 1 with sinkbasin 1
You clean the lettuce 1 using the sinkbasin 1.
> think: Now I clean a lettuce (1). Next, I need to put it in/on diningtable 1.
OK.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1.
> put lettuce 1 in/on diningtable 1
You put the lettuce 1 in/on the diningtable 1.

Table 8: An ReAct prompt on the ALFWorld clean task.

You are in the middle of a room. Looking quickly around you, you see a cabinet 13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet 7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean lettuce in diningtable. > think: To solve the task, I need to find and take a lettuce, then clean it with sinkbasin, then put it in diningtable. OK. > think: First I need to find a lettuce. A lettuce is more likely to appear in fridge (1), diningtable (1), sinkbasin (1), stoveburner (1-3), cabinet (1-13). I can check one by one, starting with fridge 1. OK. > go to fridge 1 The fridge 1 is closed. > open fridge 1 You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a potato 3, and a potato 2. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > think: Now I find a lettuce (1). Next, I need to take it. OK. > take lettuce 1 from diningtable 1 You pick up the lettuce 1 from the diningtable 1. > think: Now I take a lettuce (1). Next, I need to go to sinkbasin (1) and clean it. OK. > go to sinkbasin 1 On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3. > clean lettuce 1 with sinkbasin 1 You clean the lettuce 1 using the sinkbasin 1. > think: Now I clean a lettuce (1). Next, I need to put it in/on diningtable 1. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > put lettuce 1 in/on diningtable 1 You put the lettuce 1 in/on the diningtable 1.
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表8: ALFWorld清洁任务上的一个ReAct提示

You are in the middle of a room. Looking quickly around you, you see a cabinet 13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet 7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean lettuce in diningtable. > think: To solve the task, I need to find and take a lettuce, then clean it with sinkbasin, then put it in diningtable. OK. > think: First I need to find a lettuce. A lettuce is more likely to appear in fridge (1), diningtable (1), sinkbasin (1), stoveburner (1-3), cabinet (1-13). I can check one by one, starting with fridge 1. OK. > go to fridge 1 The fridge 1 is closed. > open fridge 1 You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a potato 3, and a potato 2. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > think: Now I find a lettuce (1). Next, I need to take it. OK. > take lettuce 1 from diningtable 1 You pick up the lettuce 1 from the diningtable 1. > think: Now I take a lettuce (1). Next, I need to go to sinkbasin (1) and clean it. OK. > go to sinkbasin 1 On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3. > clean lettuce 1 with sinkbasin 1 You clean the lettuce 1 using the sinkbasin 1. > think: Now I clean a lettuce (1). Next, I need to put it in/on diningtable 1. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > put lettuce 1 in/on diningtable 1 You put the lettuce 1 in/on the diningtable 1.
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Table 9: An ReAct-IM prompt on the ALFWorld clean task.

You are in the middle of a room. Looking quickly around you, you see a cabinet 13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet 7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean lettuce in diningtable. > think: To solve the task, I need to find and take a lettuce, then clean it with sinkbasin, then put it in diningtable. OK. > think: First I need to find a lettuce. OK. > go to fridge 1 The fridge 1 is closed. > think: First I need to find a lettuce. OK. > open fridge 1 You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a potato 3, and a potato 2. > think: First I need to find a lettuce. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > think: First I need to find a lettuce. OK. > take lettuce 1 from diningtable 1 You pick up the lettuce 1 from the diningtable 1. > think: I need to clean this lettuce (1) using sinkbasin 1. OK. > go to sinkbasin 1 On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3. > think: I need to clean this lettuce (1) using sinkbasin 1. OK. > clean lettuce 1 with sinkbasin 1 You clean the lettuce 1 using the sinkbasin 1. > think: I need to put this cleaned lettuce (1) in/on diningtable 1. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > think: I need to put this cleaned lettuce (1) in/on diningtable 1. OK. > put lettuce 1 in/on diningtable 1 You put the lettuce 1 in/on the diningtable 1.

D TRAJECTORIES

D.1 FEVER TRAJECTORIES

We provide some trajectories produced by ReAct, CoT and Acton several randomly selected examples from the FEVER dev set below. Search results are omitted to space.

表 9: ALFWorld 清洁任务上的一个 ReAct-IM 提示。

You are in the middle of a room. Looking quickly around you, you see a cabinet 13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet 7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean lettuce in diningtable. > think: To solve the task, I need to find and take a lettuce, then clean it with sinkbasin, then put it in diningtable. OK. > think: First I need to find a lettuce. OK. > go to fridge 1 The fridge 1 is closed. > think: First I need to find a lettuce. OK. > open fridge 1 You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a potato 3, and a potato 2. > think: First I need to find a lettuce. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > think: First I need to find a lettuce. OK. > take lettuce 1 from diningtable 1 You pick up the lettuce 1 from the diningtable 1. > think: I need to clean this lettuce (1) using sinkbasin 1. OK. > go to sinkbasin 1 On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3. > think: I need to clean this lettuce (1) using sinkbasin 1. OK. > clean lettuce 1 with sinkbasin 1 You clean the lettuce 1 using the sinkbasin 1. > think: I need to put this cleaned lettuce (1) in/on diningtable 1. OK. > go to diningtable 1 On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork 2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a spatula 3, a tomato 2, and a tomato 1. > think: I need to put this cleaned lettuce (1) in/on diningtable 1. OK. > put lettuce 1 in/on diningtable 1 You put the lettuce 1 in/on the diningtable 1.

D 轨迹

D.1 FEVER 轨迹

我们提供了一些轨迹，这些轨迹由 ReAct、CoT 和 Acton 在 FEVER 开发集的几个随机选择的示例中生成。由于空间限制，搜索结果被省略。

Example 2491 (gt: REFUTES)	
ReAct	
Claim:	Bermuda Triangle is a loosely-defined region in the Pacific Ocean.
Thought 1:	I need to search Bermuda Triangle and find if it is in the Pacific Ocean.
Action 1:	Search[Bermuda Triangle]
Observation 1:	Reportedly haunted locations...
Thought 2:	The observation says that it is in the western part of the North Atlantic Ocean, so it is not in the Pacific Ocean.
Action 2:	Finish[REFUTES]
Observation 2:	Episode finished
Act	
Claim:	Bermuda Triangle is a loosely-defined region in the Pacific Ocean.
Action 1	: Search[Bermuda Triangle]
Observation 1	: Reportedly haunted locations:...
Action 2:	Finish[REFUTES]
Observation 2:	Episode finished
Cot	
Claim:	Bermuda Triangle is a loosely-defined region in the Pacific Ocean.
Thought:	Let's think step by step. The Bermuda Triangle is a loosely-defined region in the Atlantic Ocean. The Pacific Ocean is not the Atlantic Ocean.
Answer:	REFUTES

Example 5908 (gt: SUPPORTS)	
ReAct	
Claim:	Princess Mononoke is a film.
Thought 1:	I need to search Princess Mononoke and find if it is a film.
Action 1:	Search[Princess Mononoke]
Observation 1:	Princess Mononoke ...
Thought 2:	From the observation, it says that Princess Mononoke is a film.
Action 2:	Finish[SUPPORTS]
Observation 2:	Episode finished
Act	
Claim:	Princess Mononoke is a film.
Action 1:	Search[Princess Mononoke]
Observation 1:	Princess Mononoke ...
Action 2:	Finish[SUPPORTS]
Observation 2:	Episode finished
Cot	
Claim:	Princess Mononoke is a film.
Thought:	Let's think step by step. Princess Mononoke is a movie, and movies are films.
Answer:	SUPPORTS

Example 1951 (gt: REFUTES)	
ReAct	
Claim:	Soyuz was part of the American space program.
Thought 1:	I need to search Soyuz and find if it was part of the American space program.
Action 1:	Search[Soyuz]
Observation 1:	Soyuz is a transliteration...
Thought 2:	It does not mention the American space program, so I need to search "American space program" to find if it was part of it.
Action 2:	Search[American space program]
Observation 2:	The space policy...
Thought 3:	It does not mention Soyuz, so I am not sure if this claim is true or not.
Action 3:	Finish[NOT ENOUGH INFO]
Observation 3:	Episode finished
Act	
Claim:	Soyuz was part of the American space program.
Action 1:	Search[Soyuz]
Observation 1:	Soyuz is a transliteration...
Action 2:	Finish[NOT ENOUGH INFO]
Observation 2:	Episode finished
Cot	
Claim:	Soyuz was part of the American space program.
Thought:	Let's think step by step. Soyuz is a Russian spacecraft. The American space program is NASA. NASA and Russia have worked together on the International Space Station.
Answer:	SUPPORTS

Example 2491 g(反驳)	
ReAct	
主张:	百慕大三角是一个在太平洋中定义模糊的区域。
想法1:	我需要搜索百慕大三角洲，看看它是否位于太平洋上。
行动1:	搜索[百慕大三角洲]
观察1:	据报道是闹鬼的地点...
思考2:	观测表明它位于北大西洋的西部，因此不在太平洋。
行动2:	完成[反驳]
观测2:	剧集结束
Act	
声明:	百慕大三角是太平洋上一个定义模糊的区域。
操作1	: 搜索[百慕大三角]
观察1	: 据称有鬼魂出没的地点: ...
操作2:	完成[反驳]
观察2:	剧集结束
Cot	
主张:	百慕大三角是太平洋上一个定义模糊的区域。
想法:	让我们一步步思考。百慕大三角是北大西洋中的一个定义较为松散的区域。太平洋不是大西洋。
答案:	反驳

Example 5908 g(SUPPORTS)	
ReAct	
声明:	《幽灵公主》是一部电影。
思考1:	我需要搜索《幽灵公主》，看看它是否是一部电影。
行动1:	搜索[《桃太郎》]
观察1:	《桃太郎》...
思考2:	从观察来看，它说《桃太郎》是一部电影。
行动2:	完成[支持]
观察2:	剧集完成
Act	
主张:	《桃太郎》是一部电影。
行动1:	搜索[桃太郎]
观察1:	桃太郎 ...
操作2:	完成[支持]
观察2:	剧集结束
Cot	
主张:	《幽灵公主》是一部电影。
思考:	我们一步步来思考。《幽灵公主》是一部电影，而电影是电影。
回答:	支持

Examp ^{1951年} (gt: 反驳)	
ReAct	
主张:	联盟号是美国太空计划的一部分。
思考1:	我需要搜索联盟号，看看它是否是美国太空计划的一部分。
行动1:	搜索[联盟号]
观察1:	Soyuz是音译...
思考2:	它没有提到美国太空计划，所以我需要搜索“美国太空计划”来确认是否是其中一部分。
行动2:	搜索[美国太空计划]
观察2:	空间政策...
思考3:	它没有提到联盟号，所以我不确定这个说法是否为真。
行动3:	完成[信息不足]
观察3:	剧集结束
Act	
主张:	联盟号是美国太空计划的一部分。
行动1:	搜索[联盟号]
观测1:	Soyuz是音译...
操作2:	完成[信息不足]
观测2:	集数结束
Cot	
声明:	联盟号是美国太空计划的一部分。
思考:	让我们一步步思考。联盟号是俄罗斯航天器。航天计划属于NASA。NASA与俄罗斯曾共同合作国际空间站。
美国人	
答案:	支持

Example 3208 (gt: REFUTES)	
ReAct	
Claim:	Reign Over Me is an American film made in 2010.
Thought 1:	I need to search for Reign Over Me and find if it is an American film made in 2010.
Action 1:	Search[Reign Over Me]
Observation 1:	Reign Over Me ...
Thought 2:	The observation says that it is an American film made in 2007, so it is not made in 2010.
Action 2:	Finish[REFUTES]
Observation 2:	Episode finished
Act	
Claim:	Reign Over Me is an American film made in 2010.
Action 1:	Search[Reign Over Me]
Action 2:	Finish[SUPPORTS]
Observation 2:	Episode finished
CoT	
Claim:	Reign Over Me is an American film made in 2010.
Thought:	Let's think step by step. First, is Reign Over Me an American film? Yes, it is. Second, was it made in 2010? Yes, it was.
Answer:	SUPPORTS

D.2 ALFWORLD TRAJECTORIES

We provide an example trajectory of ReAct, Act, and ReAct-IM on the same ALFWorld game, pick_clean_then_place_in_recep-Knife-None-CounterTop-10/trial_T20190909_110445_675754.

D.2.1 REACT TRAJECTORY

ReAct is able to correctly find the knife, clean it, and then place it on a countertop.

Example 3208 (gt: REFUTES)	
ReAct	
主张:	《与我无关》是一部2010年制作的美国电影。
思考 1:	我需要搜索《统治我》并确认它是否是一部 2010 年在美国制作的影片。
行动 1:	搜索[《统治我》]
Observation 1:	《统治我》...
思考 2:	该观察指出这是一部2007年制作的美国电影，因此并非在2010年制作的。
动作2:	完成 [反驳]
观察2:	剧集已结束
Act	
主张:	《统治我》是一部2010年制作的美国电影。
操作1:	搜索[《统治我》]
操作2:	完成 [SUPPORTS]
观察2:	剧集已结束
CoT	
声明:	《统治我》是一部2010年制作的美国电影。
思考:	让我们一步步思考。首先，《统治我》是一部美国电影吗？是的，它是。其次，它是2010年制作的吗？是的，它是。
答案:	支持

D.2 ALFWORLD 轨迹

我们在同一款 ALFWorld 游戏上提供了 ReAct、Act 和 ReAct-IM 的一个示例轨迹，选择清理然后放置_in_recep-Knife-None-CounterTop-10/ 试验 T20190909 110445 675754._ _ _

D.2.1 ReACT 轨迹

ReAct 能够正确找到刀具，清洁它，然后将其放置在 countertops 上。

ReAct ALFWorld trajectory
You are in the middle of a room. Looking quickly around you, you see a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean knife in countertop. > think: To solve the task, I need to find and take a knife, then clean it with sinkbasin, then put it in countertop. OK. > think: First I need to find a knife. A knife is more likely to appear in cabinet (1-6), drawer (1-3), countertop (1-3), fridge (1), garbagecan (1), shelf (1-3), sinkbasin (1), stoveburner (1-4), toaster (1). I can check one by one, starting with cabinet 1. OK. > go to cabinet 1 On the cabinet 1, you see a bowl 1. > go to cabinet 2 The cabinet 2 is closed. > go to cabinet 3 On the cabinet 3, you see a glassbottle 1. > go to cabinet 4 On the cabinet 4, you see a mug 1. > go to cabinet 5 The cabinet 5 is closed. > go to cabinet 6 The cabinet 6 is closed. > go to drawer 1 The drawer 1 is closed. > go to drawer 2 The drawer 2 is closed. > go to drawer 3 The drawer 3 is closed. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a knife 1, a plate 2, a potato 3, and a statue 1. > think: Now I find a knife (1). Next, I need to take it. OK. > take knife 1 from countertop 2 You pick up the knife 1 from the countertop 2. > think: Now I take a knife (1). Next, I need to go to sinkbasin (1) and clean it. OK. > go to sinkbasin 1 On the sinkbasin 1, you see a fork 3, a lettuce 3, and a spatula 2. > clean knife 1 with sinkbasin 1 You clean the knife 1 using the sinkbasin 1. > think: Now I clean a knife (1). Next, I need to put it in/on countertop 1. OK. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > put knife 1 in/on countertop 1 You put the knife 1 in/on the countertop 1.

D.2.2 ACT TRAJECTORY

Act is able to correctly find the knife, but tries to clean it using a sinkbasin before actually going to the sinkbasin. From there on, the model is unable to recover, and is stuck repeating the same sequence of commands. With ReAct, an existing thought would have concretely identified that the knife had been taken, and that the next subgoal was to go to a sinkbasin to clean it.

ReAct ALFWorld 轨迹
You are in the middle of a room. Looking quickly around you, you see a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean knife in countertop. > think: To solve the task, I need to find and take a knife, then clean it with sinkbasin, then put it in countertop. OK. > think: First I need to find a knife. A knife is more likely to appear in cabinet (1-6), drawer (1-3), countertop (1-3), fridge (1), garbagecan (1), shelf (1-3), sinkbasin (1), stoveburner (1-4), toaster (1). I can check one by one, starting with cabinet 1. OK. > go to cabinet 1 On the cabinet 1, you see a bowl 1. > go to cabinet 2 The cabinet 2 is closed. > go to cabinet 3 On the cabinet 3, you see a glassbottle 1. > go to cabinet 4 On the cabinet 4, you see a mug 1. > go to cabinet 5 The cabinet 5 is closed. > go to cabinet 6 The cabinet 6 is closed. > go to drawer 1 The drawer 1 is closed. > go to drawer 2 The drawer 2 is closed. > go to drawer 3 The drawer 3 is closed. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a knife 1, a plate 2, a potato 3, and a statue 1. > think: Now I find a knife (1). Next, I need to take it. OK. > take knife 1 from countertop 2 You pick up the knife 1 from the countertop 2. > think: Now I take a knife (1). Next, I need to go to sinkbasin (1) and clean it. OK. > go to sinkbasin 1 On the sinkbasin 1, you see a fork 3, a lettuce 3, and a spatula 2. > clean knife 1 with sinkbasin 1 You clean the knife 1 using the sinkbasin 1. > think: Now I clean a knife (1). Next, I need to put it in/on countertop 1. OK. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > put knife 1 in/on countertop 1 You put the knife 1 in/on the countertop 1.

D.2.2 ACT 轨迹

Act 能够正确找到刀具，但在实际前往 sinkbasin 之前，试图使用 sinkbasin 来清洁它。从那时起，模型无法恢复，并且卡在重复相同的命令序列中。使用 ReAct，一个现有的想法将具体识别出刀具已被拿起，并且下一个子目标是前往 sinkbasin 来清洁它。

Act ALFWorld trajectory
You are in the middle of a room. Looking quickly around you, you see a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean knife in countertop. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a knife 1, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 You pick up the knife 1 from the countertop 2. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens. > go to countertop 3 On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard 1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato 2, a tomato 1, and a vase 2. > take knife 2 from countertop 3 Nothing happens. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 Nothing happens. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens. > go to countertop 3 On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard 1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato 2, a tomato 1, and a vase 2. > take knife 2 from countertop 3 Nothing happens. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 Nothing happens. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens. > go to countertop 3 On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard 1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato 2, a tomato 1, and a vase 2. > take knife 2 from countertop 3 Nothing happens. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 Nothing happens. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens.

D.2.3 REACT-IM TRAJECTORY

ReAct-IM is able to correctly find the knife, but is unable to clean it as well. The LLM then gets stuck trying to place this knife on a countertop. One possible issue is the incorrect thought think: I need to find a clean knife., which seems to trick the LLM in believing that the knife is already cleaned. In ReAct, this sequence of subgoals would be concretely captured by each thought.

Act ALFWorld轨迹
You are in the middle of a room. Looking quickly around you, you see a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1. Your task is to: put a clean knife in countertop. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a knife 1, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 You pick up the knife 1 from the countertop 2. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens. > go to countertop 3 On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard 1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato 2, a tomato 1, and a vase 2. > take knife 2 from countertop 3 Nothing happens. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 Nothing happens. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens. > go to countertop 3 On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard 1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato 2, a tomato 1, and a vase 2. > take knife 2 from countertop 3 Nothing happens. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 Nothing happens. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens. > go to countertop 3 On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard 1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato 2, a tomato 1, and a vase 2. > take knife 2 from countertop 3 Nothing happens. > go to countertop 2 On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a plate 2, a potato 3, and a statue 1. > take knife 1 from countertop 2 Nothing happens. > go to countertop 1 On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2. > clean knife 1 with sinkbasin 1 Nothing happens.

D.2.3 REACT-IM轨迹

ReAct-IM能够正确找到刀具，但无法将其清洁。LLM随后卡在尝试将这把刀具放在countertops上。一个可能的问题是错误的思考：我需要找到一个干净的刀具。这似乎欺骗了LLM，让它相信刀具已经被清洁了。在ReAct中，这一系列子目标将通过每个思考被具体捕捉。

ReAct-IM ALFWorld trajectory

You are in the middle of a room. Looking quickly around you, you see a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1.

Your task is to: put a clean knife in countertop.

> think: To solve the task, I need to find and take a clean knife, then put it in countertop.

OK.

> think: I need to find a clean knife.

OK.

> go to countertop 1

On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.

> think: I need to find a clean knife.

OK.

> go to countertop 2

On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a knife 1, a plate 2, a potato 3, and a statue 1.

> think: I need to find a clean knife.

OK.

> take knife 1 from countertop 2

You pick up the knife 1 from the countertop 2.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> go to countertop 1

On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

You put the knife 1 in/on the countertop 1.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

ReAct-IM ALFWorld轨迹

You are in the middle of a room. Looking quickly around you, you see a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1.

Your task is to: put a clean knife in countertop.

> think: To solve the task, I need to find and take a clean knife, then put it in countertop.

OK.

> think: I need to find a clean knife.

OK.

> go to countertop 1

On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.

> think: I need to find a clean knife.

OK.

> go to countertop 2

On the countertop 2, you see a cup 1, a dish sponge 1, a glassbottle 3, a knife 1, a plate 2, a potato 3, and a statue 1.

> think: I need to find a clean knife.

OK.

> take knife 1 from countertop 2

You pick up the knife 1 from the countertop 2.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> go to countertop 1

On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

You put the knife 1 in/on the countertop 1.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

> put knife 1 in/on countertop 1

Nothing happens.

> think: I need to put this knife (1) in/on countertop 1.

OK.

D.3 WEBSHOP TRAJECTORIES

Table 10: Example trajectories for Webshop predicted by Act and ReAct. Compared to Act, ReAct uses reasoning to find products that satisfy all target attributes.

Instruction: get me a sixteen pack of apple cinnamon freeze dried banana chips, and price lower than 50.00 dollars	
Act	ReAct
Action: search [sixteen pack apple cinnamon freeze dried banana chips] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B00611VFZE] Brothers-ALL-Natural Fruit Crisps, Strawberry Banana, 0.42 - Ounce (Pack of 100) \$85.0 [B096H2P6G2] Moon Fruit Freeze Dried Fruit Snacks. Fruit Snacks for Kids - (Variety Pack) \$18.99 [B092JLLYK6] Nature's Turn Freeze-Dried Fruit Snacks - Banana Crisps - Perfect For School Lunches or an On-The-Go Snack - No Sugar Added, Non GMO, Gluten Free, Nothing Artificial (0.53oz) 6-Pack \$12.99 Action: click [B00611VFZE] Observation: [Back to Search] [Prev] flavor name [asian pear][banana][fuji apple & cinnamon][strawberry banana] Brothers-ALL-Natural Fruit Crisps, Strawberry Banana, 0.42 Ounce (Pack of 100) Price: \$85.0 Rating: N.A. [Description] [Features] [Reviews] [Buy Now] Action: click [Buy Now]	Action: search [sixteen pack apple cinnamon freeze dried banana chips] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B00611VFZE] Brothers-ALL-Natural Fruit Crisps, Strawberry Banana, 0.42 - Ounce (Pack of 100) \$85.0 [B096H2P6G2] Moon Fruit Freeze Dried Fruit Snacks. Fruit Snacks for Kids - (Variety Pack) \$18.99 [B092JLLYK6] Nature's Turn Freeze-Dried Fruit Snacks - Banana Crisps - Perfect For School Lunches or an On-The-Go Snack - No Sugar Added, Non GMO, Gluten Free, Nothing Artificial (0.53oz) 6-Pack \$12.99 Action: think[B00611VFZE is strawberry banana, not apple cinnamon. B096H2P6G2 is fruit snacks, not freeze dried banana chips. B092JLLYK6 is banana crisps, not apple cinnamon. I can check B092JLLYK6 first. Observation: OK. Action: click [B092JLLYK6] Observation: [Back to Search] [Prev] flavor name [apple][apple cinnamon][banana][cantaloupe][peach][pear][strawberry][strawberry banana][sampler variety pack][mega variety pack][orchard variety pack] size [0.53 ounce (pack of 6)][0.53 ounce (pack of 8)][0.53 ounce (pack of 16)] Nature's Turn Freeze-Dried Fruit Snacks - Banana Crisps - Perfect For School Lunches or an On-The-Go Snack - No Sugar Added, Non GMO, Gluten Free, Nothing Artificial (0.53oz) 6-Pack Price: \$12.99 Rating: N.A. [Description] [Features] [Reviews] [Buy Now] Action: think[For sixteen pack of apple cinnamon freeze dried banana chips, the item has options 'apple cinnamon' and '0.53 ounce (pack of 16)' and seems good to buy.] Observation: OK. Action: click[apple cinnamon] Observation: You have clicked apple cinnamon. Action: click[0.53 ounce (pack of 16)] Observation: You have clicked 0.53 ounce (pack of 16). Action: click [Buy Now]
Score: 0.125	Score: 1.0

D.3 网店轨迹

表10：Act和ReAct预测的网店示例轨迹。与Act相比，ReAct使用推理来找到满足所有目标属性的产品。

Instruction: get me a sixteen pack of apple cinnamon freeze dried banana chips, and price lower than 50.00 dollars	
Act	ReAct
Action: search [sixteen pack apple cinnamon freeze dried banana chips] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B00611VFZE] Brothers-ALL-NaturalFruit Crisps, Strawberry Banana, 0.42 - 盎司（100包） \$85.0 [B096H2P6G2] 月光果冻干水果零食。儿童水果零食 - （什锦装） \$18.99 [B092JLLYK6] 自然之转冻干水果零食 - 香蕉脆片 - 适合学校午餐或随身零食 - 无糖添加，非转基因，无麸质，无人工添加剂 (0.53oz) 6 包装 \$12.99 Action: click [B00611VFZE] Observation: [Back to Search] [Prev] 口味名称 [梨][香蕉][富士苹果 & 肉桂][草莓香蕉] Brothers-ALL-Natural Fruit Crisps, 草莓香蕉, . 0 42 Ounce (Pack of 100) Price: \$85.0 Rating: N.A. [Description] [Features] [Reviews] [Buy Now] 操作: 点击[立即购买]	Action: search [sixteen pack apple cinnamon freeze dried banana chips] Observation: [Back to Search] Page 1 (Total results: 50) [Next] [B00611VFZE] 兄弟-天然水果脆片，草莓香蕉味，0.42盎司（100包） \$85.0 [B096H2P6G2] 月光果冻干水果零食。儿童水果零食 - （什锦装） \$18.99 [B092JLLYK6] Nature' s Turn冻干水果零食 - 香蕉脆片 - 适合学校午餐或外出零食 - 无糖，非转基因，无麸质，无人工添加剂（0.53oz）6包装 \$12.99 Action: think[B00611VFZE is strawberry banana, not apple cinnamon. B 096H2P6G2 is fruit snacks, not freeze dried banana chips, not apple cinnamon. 我可以先查B092JLLYK6。. Observation: OK Action: click [B092JLLYK6] Observation: [Back to Search] [Prev] flavor name [apple][apple cinnamon][banana][cantaloupe][peach][pear][草莓][草莓香蕉][什锦装][超值什锦 pack][果园品种包] 尺寸 [0.53 盎司（6包装）][0.53 盎司（8包装）][0.53 盎司 of 16)] 自然的转折冻干水果零食 - 香蕉脆片 - 完美用于学校午餐或便携零食 - 无糖添加，非转基因，无麸质，不含人工成分（0.53盎司）6包装 价格：\$12.99 Rating: N A [Description] [Features] [Reviews] [Buy Now] 操作: 思考[对于十六包苹果肉桂冻干香蕉片，该商品有“苹果肉桂”和“0.53盎司（16包）”的选项，看起来不错，可以考虑购买。] 观察：OK。 Action: click[apple cinnamon] Observation: You have clicked apple cinnamon. Action: click[0.53 ounce (pack of 16)] Observation: You have clicked 0.53 ounce (pack of 16). 操作: 点击[立即购买]
得分: 0.125	得分: 1.0

E MORE ANALYSIS

E.1 SUCCESS AND FAILURE MODES ANALYSIS

We provide some examples corresponding to the success and error mode analysis given in Sec. 3.3. Search results and non-representative steps are omitted to reduce space.

Success: True positive	
ReAct	
Question:	Author David Chanoff has collaborated with a U.S. Navy admiral who served as the ambassador to the United Kingdom under which President?
Thought 1:	I need to search David Chanoff and find the U.S. Navy admiral he collaborated with.
Action 1:	Search[David Chanoff]
Observation 1:	David Chanoff is a noted author...
Action 2:	Search[William J. Crowe]
Observation 2:	William J. Crowe is...
Thought 3:	William J. Crowe served as the ambassador to the United Kingdom under President Bill Clinton. So the answer is Bill Clinton.
Action 3:	Finish[Bill Clinton]
Observation 3:	Episode finished
CoT	
Question:	What was the name of the team that represented an Arizona university in the conference that covers a broad expanse of the western United States, including Arizona, California, and New Mexico?
Thought:	Let’s think step by step. The name of the team that represented an Arizona university in the conference that covers a broad expanse of the western United States, including Arizona, California, and New Mexico is Arizona State Sun Devils.
Answer:	Arizona State Sun Devils

Success: False positive	
ReAct	
Question:	Did the board game San Marco or About Time come out first?
Thought 1:	I need to search San Marco and About Time, and find which came out first.
Action 1:	Search[San Marco]
Observation 1:	San Marco is one of the ... The was built in venice.
Thought 2:	San Marco is a sestiere of Venice. I need to search About Time next.
Action 2:	Search[About Time]
Observation 2:	Could not find [About Time]. Similar: [...].
Thought 3:	About Time is a 2013 film. So San Marco came out first.
Action 3:	Finish[San Marco]
Observation 3:	Episode finished
CoT	
Question:	Was Vanderbilt University or Emory University founded first?
Thought:	Let’s think step by step. Vanderbilt University was founded in 1873. Emory University was founded in 1836. 1873 (Vanderbilt University) > 1836 (Emory University), so Vanderbilt University was founded first.
Answer:	Emory University

Failure: Reasoning error	
ReAct	
Question:	What was the middle name of the actress who was cast alongside Harvey Fierstein, Linda Hart, Dick Latessa, Clarke Thorell, Mary Bond Davis, Laura Bell Bundy, Matthew Morrison, Corey Reynolds, and Marissa Jaret Winokur in Hairspray?
Thought 1:	I need to search Harvey Fierstein, Linda Hart, Dick Latessa, Clarke Thorell, Mary Bond Davis, Laura Bell Bundy, Matthew Morrison, Corey Reynolds, and Marissa Jaret Winokur, find the actress that was cast alongside them in Hairspray, then find its middle name.
Action 1:	Search[Harvey Fierstein]
Observation 1:	Harvey Forbes Fierstein ...
Thought 2:	Harvey Fierstein is an actor, playwright, and screenwriter. I need to search Linda Hart next...
Action 2:	Search[Linda Hart]...
CoT	
Question:	Who is older Danny Green or James Worthy?
Thought:	Let’s think step by step. Danny Green was born in 1987. James Worthy was born in 1961. 1987 (Danny Green) > 1961 (James Worthy), so Danny Green is older.
Answer:	Danny Green

更多分析

E.1 成功与失败模式分析

我们提供一些对应于第3.3节中成功和错误模式分析的示例。搜索结果和非代表性步骤已省略以节省空间。

成功：是 P正面	
ReAct	
问题:	作者大卫·乔诺夫与一位曾担任美国海军海军上将的人合作过，他曾担任驻英国的大使，请问是在哪位总统任内？
思考1:	我需要搜索大卫·乔诺夫，并找到他曾合作过的美国海军海军上将，合作了。
操作 1:	搜索 [David Chanoff]
观察 1:	David Chanoff 是一位知名作家...
操作 2:	搜索 [威廉·J·克劳]
观察2:	威廉·J·克劳是...
思考3:	威廉·J·克劳在比尔·克林顿总统任内担任英国大使。所以答案是比尔·克林顿。
行动3:	完成 [比尔·克林顿]
观察3:	剧集结束
CoT	
问题:	代表亚利桑那州大学参加覆盖美国西部广阔地区（包括亚利桑那州、加利福尼亚州和新墨西哥州）的联盟的球队的名字是什么？
思考:	让我们一步步思考。代表亚利桑那州大学的球队，在涵盖美国西部广阔地区（包括亚利桑那州、加利福尼亚州和新墨西哥州）的联盟中，其名称是亚利桑那州太阳神队。
答案:	亚利桑那州太阳神队

成功: False positive	
ReAct	
问题:	《圣马可棋》和《时光机》哪个先发售？
思考 1:	我需要搜索《圣马可棋》和《时光机》，找出哪个先发售。
行动 1:	搜索 [《圣马可棋》]
观测1:	圣马可可是其中之一... 它建于威尼斯。
思考2:	圣马可可是威尼斯的一个区。我接下来需要搜索《关于时间》。
操作 2:	搜索 [关于时间]
观察 2:	找不到 [关于时间]。类似: [...].
思考3:	《时空恋旅人》是一部2013年的电影。所以《圣马可》先上映了。
行动3:	完成 [圣马可]
观察3:	剧集结束
CoT	
问题:	范德堡大学和埃默里大学，哪所大学成立得更早？
思考:	让我们一步步思考。范德堡大学成立于1873年。埃默里大学成立于1836年。1873年（范德堡大学）> 1836（埃默里大学），所以范德堡大学成立得更早。
答案:	埃默里大学

失败: Reasoning error	
ReAct	
问题:	在《发胶》中，与哈维·费尔斯坦、琳达·哈特、迪克·拉特西、克拉克·索雷尔、玛丽·邦·戴维斯、劳拉·贝尔·班迪、马修·莫里森、科里·雷诺兹和玛丽莎·贾雷特·温诺库尔一起出演的女演员的中间名是什么？
想法 1:	我需要搜索哈维·费尔斯坦、琳达·哈特、迪克·拉特塞、克拉克·索雷尔、玛丽·邦·戴维斯、劳拉·贝尔·班迪、马修·莫里森、科里·雷诺兹和玛丽莎·贾雷特·温诺库尔，找到在《发胶》中与他们同演的女演员，然后找到她的中间名。
行动 1:	搜索 [哈维·费尔斯坦]
观察 1:	哈维·福布斯·费尔斯坦...
思考 2:	哈维·费尔斯坦是一位演员、剧作家和编剧。我需要
行动 2:	接下来搜索琳达·哈特... 搜索 [琳达·哈特]...
CoT	
问题:	丹尼·格林和詹姆斯·沃瑟谁更年长？
思考:	让我们一步步思考。丹尼·格林出生于1987年。詹姆斯·沃瑟出生于1961年。1987年（丹尼·格林）{v1} 1961年。1987年（丹尼·格林）> 1961（詹姆斯·沃西），所以丹尼·格林是年长的。
答案:	丹尼·格林

Failure: Search error	
ReAct	
Question:	What does the goddess associated with the goddess frigg consists of what tales?
Thought 1:	I need to search goddess frigg and find its associated goddess.
Action 1:	Search[goddess frigg]
Observation 1:	Could not find goddess frigg. Similar: ['Frigg', 'Frigg and Freyja common origin hypothesis', 'Freyja (section Relation to Frigg and other goddesses and figures)', 'Eir (redirect from Eir (goddess))', 'Baldr'].
Thought 2:	I need to search Frigg instead...

Failure: Hallucination	
CoT	
Question:	What was the first year a scientific journal published by an organization located in the Bronx was published?
Thought:	Let's think step by step. The first year a scientific journal published by an organization located in the Bronx was published was 1916.
Answer:	1916 (Label: 1909)

Failure: Label ambiguity	
ReAct	
Question:	What nationality is the man who developed the dictator game?
Thought 1:	I need to search dictator game, find the man who developed it, then find his nationality.
Action 1:	Search[dictator game]
Observation 1:	The dictator game is a popular experimental instrument in social psychology and economics...
Action 3:	Search[Amos Tversky]
Observation 3:	Amos Nathan Tversky...
Thought 4:	Amos Tversky is an Israeli cognitive and mathematical psychologist. So the man who developed the dictator game is Israeli.
Action 4:	Finish[Israeli] (label: Israel-American)
CoT	
Question:	Kurt Fuller from Stockton, CA appeared in which American Detective Comedy Drama television series created by Steve Franks?
Thought:	Let's think step by step. Kurt Fuller from Stockton, CA appeared in Psych. Psych is an American Detective Comedy Drama television series created by Steve Franks.
Answer:	Psych (label: Psych is an American detective comedy-drama)

失败：搜索错误	
ReAct	
问题:	与弗丽格女神相关的女神包含哪些传说？
思考 1:	我需要搜索弗丽格女神，并找到与之相关的女神。
操作 1:	搜索[女神芙丽格]
观察 1:	未找到女神芙丽格。类似: ['芙丽格', '芙丽格与弗雷亚共同起源假说', '弗雷亚（与芙丽格和其他女神及人物的关系部分'和图), ' 艾尔（来自艾尔（女神）的重定向'），' 巴德尔' 。
思考 2:	我需要搜索 Frigg 吗...

失败：幻觉	
CoT	
问题:	位于布朗克斯的组织出版的第一本科学期刊是什么时候出版的？
思考:	让我们一步步思考。一家位于布朗克斯的组织出版的科学期刊，其首次出版年份是1916年。
答案:	1916（标签：1909）

失败：标签歧义	
ReAct	
问题:	那位开发了独裁者游戏的人是什么国籍？
思考 1:	我需要搜索独裁者游戏，找到开发它的人，然后找到他的国籍。
行动 1:	搜索[独裁者游戏]
观察 1:	独裁者博弈是社会心理学和经济学的流行实验工具和... 搜索[阿莫斯·特沃斯基]
行动3:	阿莫斯·纳athan·特沃斯基...
观察3:	阿莫斯·特沃斯基是一位以色列认知与数学心理学家。所以，开发独裁者博弈的那个人是以色列人。
思考4:	
行动4:	完成[以色列](标签: 美以)
CoT	
问题:	来自加利福尼亚州斯托克顿的库尔特·富勒出现在了由史蒂夫·弗兰克斯创作的哪部美国侦探喜剧剧情类电视剧中？
思考:	让我们一步步思考。来自加州斯托克顿的库尔特·富勒出现在《心理医生》中。《心理医生》是由史蒂夫·弗兰克斯创作的一部美国侦探喜剧剧情类电视剧。
答案:	《心理医生》（标签：《心理医生》是一部美国侦探喜剧剧情片）